



US Army Corps
of Engineers®

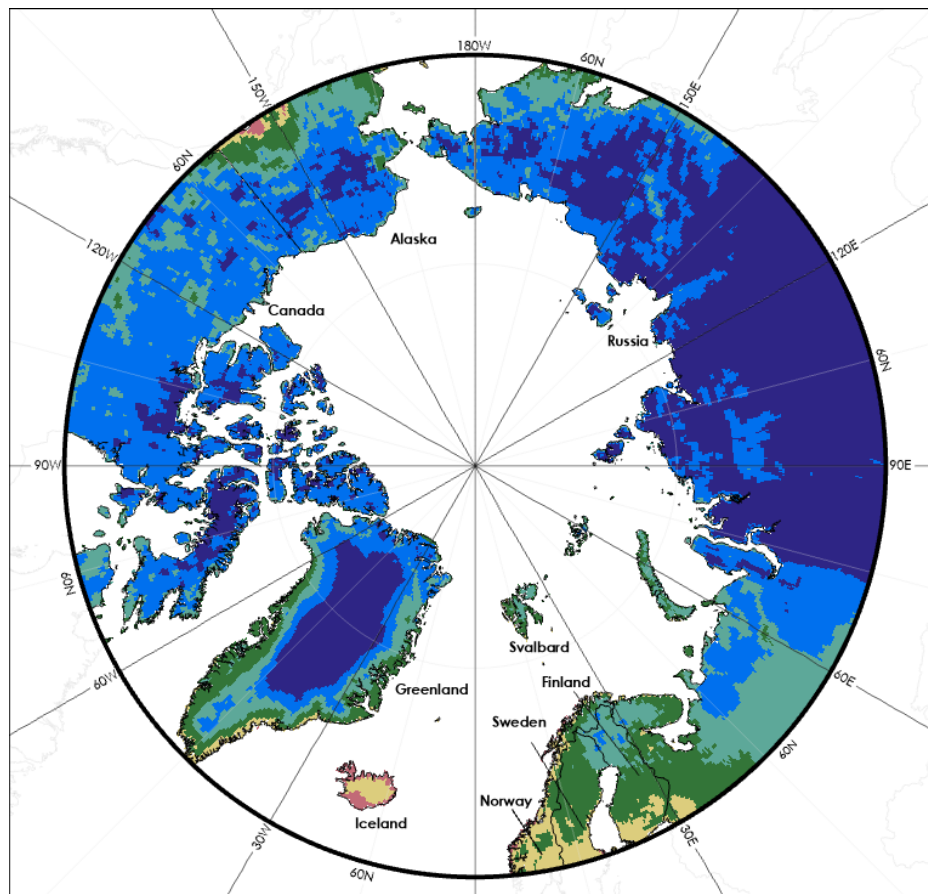


*Environmental Security Technology Certification Program (ESTCP) Alaska
Innovation Landscape Network (AILN)*

Arctic and Subarctic Zonal Characterization and Operational Thresholding (AZCOT)

Rosa T. Affleck, Michelle L. Michaels, Brendon Hoch,
Evan P. Barrett, Jenna N. Williams, Zachary J. Zody,
Paige Toebben, and Mathew E. Hefner

March 2026



The US Army Engineer Research and Development Center (ERDC) solves the nation's toughest engineering and environmental challenges. ERDC develops innovative solutions in civil and military engineering, geospatial sciences, water resources, and environmental sciences for the Army, the Department of Defense, civilian agencies, and our nation's public good. Find out more at www.erdclibrary.on.worldcat.org/discovery.

To search for other technical reports published by ERDC, visit the ERDC online library at <http://www.erdclibrary.on.worldcat.org/discovery>.

Arctic and Subarctic Zonal Characterization and Operational Thresholding (AZCOT)

Rosa T. Affleck, Michelle L. Michaels, Brendon Hoch, Jenna N. Williams, Zachary J. Zody,
Paige Toeppen, and Mathew E. Hefner

*US Army Engineer Research and Development Center (ERDC)
Cold Regions Research and Engineering Laboratory (CRREL)
72 Lyme Road
Hanover, NH 03755-1290*

Evan P. Barrett

*Oak Ridge Associated Universities (ORAU)
Oak Ridge Institute for Science and Education (ORISE)
2902 Newmark Drive
PO Box 9005
Champaign, IL 61826-9005*

Final Technical Report

Distribution Statement A. Approved for public release: distribution is unlimited.

Prepared for US Army Corps of Engineers–Headquarters
Washington, DC 20314-1000

Under Project NH23-8375, “Arctic Zonal Characterization and Operational
Thresholding (AZCOT)”

Abstract

The US military develops and updates environmental parameters specified for the sustainment of operations throughout the world. These requirements are generally based on environmental data providing a baseline of temperature, wind, and precipitation expectations for each location. Observational data for Arctic regions is limited because of the remote and sparsely occupied geographical conditions. To address the need for updating these requirements, a 30-year analysis of meteorological conditions was conducted using a European Centre for Medium-Range Weather Forecasts (ECMWF) global reanalysis dataset over the Arctic and Subarctic region, defined by latitude 60°–90° North for this project. Raw hourly datasets were acquired, and the minimum temperatures, maximum wind speeds, maximum snow depths, and averages were determined over the period 1991–2020 between the months of October and March for each parameter. These were then visualized with geospatial analysis, producing a variety of maps designed to assist with the classification of parameters in Arctic zones of operation across a range of temporal resolutions. Finally, a review of operational limits for military equipment was conducted to match northern zones of operation with suitable capabilities dependent on environmental conditions.

DISCLAIMER: The contents of this report are not to be used for advertising, publication, or promotional purposes. Citation of trade names does not constitute an official endorsement or approval of the use of such commercial products. All product names and trademarks cited are the property of their respective owners. The findings of this report are not to be construed as an official Department of the Army position unless so designated by other authorized documents.

DESTROY THIS REPORT WHEN NO LONGER NEEDED. DO NOT RETURN IT TO THE ORIGINATOR.

Contents

Abstract	ii
Preface.....	vii
1 Introduction.....	1
1.1 Background.....	1
1.2 Objective.....	5
1.3 Approach	5
2 Methodology	6
2.1 Area of Study.....	6
2.2 Data	6
2.2.1 European Center for Medium-Range Weather Forecasting (ECMWF) Reanalysis v5 (ERA5) as Baseline Gridded Dataset	7
2.2.2 Data Selection and Transformation	8
2.2.3 Point Observations Using Meteorological Aerodrome Report (METAR).....	10
2.3 Geospatial Analyses	11
2.3.1 Metric Calculations	11
2.3.2 Generation of Static Maps.....	12
3 Review of Thresholds.....	13
3.1 Zones and Thresholds	13
3.2 Arctic Military Equipment and Gear for Specific Zones	19
4 Geospatial Characterization Results	31
4.1 Temperature	31
4.1.1 Skin Temperature (SKT)	31
4.1.2 Two-meter Temperature (2T).....	34
4.1.3 Soil Temperature, Level 1 (STL1).....	37
4.1.4 Wind Chill Temperature (WCT)	40
4.2 Wind.....	42
4.3 Snow	45
4.3.1 Snow Depth (SD)	45
4.3.2 Snow Load (SL)	47
5 Summary.....	49
6 Discussion and Recommendations.....	50
6.1 Considerations for Soldiers.....	50
6.2 Deliverables and Future Needs	50
Bibliography	52
Abbreviations.....	55
Report Documentation Page (SF 298)	59

Figures and Tables

Figures

1.	Arctic and Subarctic terrain characteristics.	2
2.	Low mean temperatures for January (<i>left</i>) and February (<i>right</i>) from the US Air Force, 14th Weather Squadron. Obtained from the Climate Forecast System (CFS) v1, 1979–2010. (Images reproduced from USAF, n.d. Public domain.)	4
3.	The steps used for Arctic zonal characterization.	5
4.	Monthly average skin temperature (SKT) between 60°N and 90°N for October (A), November (B), December (C), January (D), February (E), and March (F), from hourly data between 1991 and 2020.	32
5.	Monthly lowest reanalysis hourly SKT between 60°N and 90°N latitude for October (A), November (B), December (C), January (D), February (E), and March (F), from hourly data between 1991 and 2020.	33
6.	The monthly frequency of SKT between 60°N and 90°N latitude being recorded at or less than –40°F for October (A), November (B), December (C), January (D), February (E), and March (F), from hourly data between 1991 and 2020.	34
7.	Monthly average 2 m temperature (2T) between 60°N and 90°N for October (A), November (B), December (C), January (D), February (E), and March (F), from hourly data between 1991 and 2020.	35
8.	Monthly lowest reanalysis hourly 2T between 60°N and 90°N latitude for October (A), November (B), December (C), January (D), February (E), and March (F), from hourly data between 1991 and 2020.	36
9.	The monthly frequency of 2T between 60°N and 90°N latitude being recorded at or less than –40°F for October (A), November (B), December (C), January (D), February (E), and March (F), from hourly data between 1991 and 2020.	37
10.	Monthly average STL1 between 60°N and 90°N latitude for October (A), November (B), December (C), January (D), February (E), and March (F), from hourly data between 1991 and 2020.	38
11.	Monthly lowest hourly STL1 between 60°N and 90°N for October (A), November (B), December (C), January (D), February (E), and March (F), from hourly data between 1991 and 2020.	39
12.	Monthly average WCT between 60°N and 90°N latitude for October (A), November (B), December (C), January (D), February (E), and March (F) from hourly data between 1991 and 2020.	41
13.	Monthly lowest WCT between 60°N and 90°N latitude for October (A), November (B), December (C), January (D), February (E), and March (F) from hourly data between 1991 and 2020.	41
14.	The monthly frequency of WCT between 60°N and 90°N latitude being recorded at or less than –65°F for October (A), November (B), December (C), January (D), February (E), and March (F) from hourly data between 1991 and 2020.	42
15.	Monthly average wind speed and direction between 60°N and 90°N latitude for October (A), November (B), December (C), January (D), February (E), and March (F), from hourly data between 1991 and 2020.	43
16.	Monthly highest wind speed between 60°N and 90°N latitude for October (A), November (B), December (C), January (D), February (E), and March (F), from hourly data between 1991 and 2020.	44

17.	The monthly frequency of wind speed between 60° N and 90° N latitude being recorded at or higher than 25 °F for October (A), November (B), December (C), January (D), February (E), and March (F), from hourly data between 1991 and 2020.	45
18.	Monthly average SD between 60° N and 90° N latitude for October (A), November (B), December (C), January (D), February (E), and March (F), from hourly data between 1991 and 2020.	46
19.	Monthly highest SD between 60° N and 90° N latitude for October (A), November (B), December (C), January (D), February (E), and March (F), from hourly data between 1991 and 2020.	46
20.	Monthly average SL between 60° N and 90° N latitude for October (A), November (B), December (C), January (D), February (E), and March (F), from hourly data between 1991 and 2020.	47
21.	Monthly highest SL between 60° N and 90° N latitude for October (A), November (B), December (C), January (D), February (E), and March (F), from hourly data between 1991 and 2020.	48

Tables

1.	Summary of cold condition limits as described in Army Regulation (AR) 70-38 (HQDA 2020).	3
2.	Temperature zones. (Data from HQDA and USMC 2025; US Army 2025.)	13
3.	Snow ranges for vehicle limitations. (Data from US Army and US Marine Corps 2025.)	14
4.	The wind chill temperature, in degrees Fahrenheit, from 10 m wind speed and 2 m air temperature. (Data from Nelson et al. 2002.)	15
5.	The time, in minutes, until frostbite occurs on exposed, dry cheek skin based on air temperature and wind speed conditions. (Data from Nelson et al. 2002.)	16
6.	Stoplight chart that the Combat Weather Teams in Alaska use. (Data from personnel at Fort Wainwright, pers. comm., February 2026.)	17
7.	Active wear of the Extended Cold Weather Clothing (ECWC) System. (Data from Technical Manual [TM] 10-8415-236-10; DA 2014.)	20
8.	Static wear of the ECWC System. (Data from TM 10-8415-236-10; DA 2014.)	20
9.	Operational temperature limits of worn gear.	21
10.	Equipment. (Data from TM 4-33.31; DA 2017.)	22
11.	Tents.	22
12.	Batteries. (Data from TM 4-33.31; DA 2017.)	23
13.	Electronics. (Data from TM 4-33.31; DA 2017.)	23
14.	Generators. (Data from TM 4-33.31; DA 2017.)	24
15.	Fuels. (Data from TM 4-33.31; DA 2017.)	24
16.	Engine oils, by military symbol. (Data from TM 4-33.31; DA 2017.)	25
17.	Hydraulic fluids listed by standard name then military symbol (if applicable). (Data from TM 4-33.31; DA 2017.)	25
18.	Gear lubricating oils, by military symbol. (Data from TM 4-33.31; DA 2017.)	25
19.	Grease lubricants. (Data from TM 4-33.31; DA 2017.)	26

20.	Solvents and cleaners, listed by military symbol. (Data from TM 4-33.31; DA 2017.)	26
21.	Miscellaneous liquids. (Data from TM 4-33.31; DA 2017.).....	26
22.	Weapon systems and components. (Data from TM 4-33.31; DA 2017.)	27
23.	Weapons lubricants. (Data from TM 4-33.31; DA 2017.)	27
24.	Rifle Lubrication guidelines. (Data from TM 4-33.31; DA 2017.)	28
25.	The M249, squad automatic weapon (SAW) lubrication guidelines. (Data from TM 4-33.31; DA 2017.).....	28
26.	Machine gun lubrication guidelines. (Data from TM 4-33.31; DA 2017).	28
27.	Mortar lubrication guidelines. (Data from TM 4-33.31; DA 2017.).....	29
28.	Artillery lubrication guidelines. (Data from TM 4-33.31; DA 2017.)	29
29.	Helicopters, according to Soldier experience.	29
30.	Fixed wing aircraft.	30
31.	Zonal statistics per location (within 60°N–90°N) for the average surface temperature, defined as the six-month mean (October–March), using hourly data from 1991–2020.....	31
32.	Zonal statistics per location (within 60°N–90°N) for the lowest recorded surface temperature, defined as the six-month minimum (October–March), from hourly data from 1991 to 2020.	33
33.	Zonal statistics per location (within 60°N and 90°N) for the average 2T, according to hourly data from October–March 1991–2020.	35
34.	Zonal statistics per location for the lowest recorded 2T, using hourly data from October–March, 1991–2020.	36
35.	Zonal statistics per location (within 60°N and 90°N) for the average STL1, using the hourly reanalysis data averaged over October to March, 1991–2020.....	38
36.	Zonal statistics per location (within 60°N and 90°N) for the lowest recorded STL1, using hourly data from October–March, 1991–2020.	39
37.	Zonal statistics per location (within 60°N and 90°N) for the average WCT, using hourly data averaged over October–March, 1991–2020.....	40
38.	Zonal statistics per location (within 60°N and 90°N) for the lowest recorded WCT, using hourly data from October–March, 1991–2020.	42
39.	Zonal statistics per location (within 60°N and 90°N) for the average wind speed (knots), using hourly data from October–March, 1991–2020.	43
40.	Zonal statistics per location (within 60°N and 90°N) for the highest recorded wind speed, using hourly data from October–March 1991–2020.....	44

Preface

This study was conducted for the US Army Corps of Engineers (USACE), Environmental Security Technology Certification Program (ESTCP) under Project NH23-8375, “Arctic Zonal Characterization and Operational Thresholding (AZCOT).” The technical monitor was Mr. Kevin Hiers, ESTCP Program Office, Office of the Deputy Assistant Secretary of Defense (Energy Resilience & Optimization), Washington, DC.

The work was performed by the Force Projection and Sustainment Branch and the Terrestrial and Cryospheric Sciences Branch of the Research and Engineering Division, US Army Engineer Research and Development Center–Cold Regions Research and Engineering Laboratory (ERDC-CRREL). At the time of publication, Ms. Sandra LeGrand was acting branch chief and Dr. John W. Weatherly was branch chief, respectively; and Dr. John B. Eylander was division chief. The acting deputy director of ERDC-CRREL was Dr. Kelly Swiderski, and the acting director was Dr. Edmund J. Russo, Jr.

The authors would like to thank Dr. Sally Shoop (ERDC-CRREL) and Dr. Ross Alter (ERDC-CRREL) for their technical review of this report, Mr. Kevin Bjella (ERDC-CRREL) for project guidance, and Mr. Martin Stuefer (University of Alaska Fairbanks) for assisting with technology transfer of project deliverables.

Special thanks to contributions from our stakeholders including Ms. Gail Weaver, 611th Air Operations Center, Joint Base Elmendorf-Richardson (JBER); CAPT Edward Garibay, Combined Arms Doctrine Directorate (CADD); Mr. Daniel Wozniczka, Arctic Regions Test Center; Mr. Rick Thoman, University of Alaska Fairbanks; and CAPT Corie Capalbo, Mr. William Henning, Mr. Ross Stark, Mr. Jacob Ballard, Mr. Jason Patla, and Mr. Benjamin Fulk, US Air Force 14th Weather Squadron.

LTC Joshua M. Haynes was commander of ERDC, and Dr. Beth C. Fleming was the director.

This page intentionally left blank.

1 Introduction

The *National Strategy for the Arctic Region* (DOS 2022) highlights the overarching strategic policies for US interests in the region. The DoD has also recently released an Arctic strategy (DoD 2024), as well as the individual Services with the Air Force (DAF 2020), Army (HQDA and DAF 2021), Navy (DON 2019), and other DoD entities requiring an increased presence across the Arctic for safeguarding the region. Mission planners must be assured the warfighter has full personnel support and operational facilities which can be quickly deployed and robust enough to maintain life sustainment in very harsh Arctic conditions, and are quickly recoverable with organic assets not requiring significant logistical support from lateral units. DoD planners have identified stringent environmental parameters (i.e., for extended operations down to -65°F) to be attained by infrastructure and equipment.¹ This is because of the difficult operating conditions of extreme cold regions where some equipment and materials will not function at certain environmental thresholds, and military operations have hard-stop operating specifications. Applying rigid requirements for environmental parameters throughout the Arctic and Subarctic is impractical in regions where those environmental conditions rarely occur. This project provides a method to apply appropriate environmental conditions using data analysis for specific areas within the Arctic and Subarctic regions from latitudes between 60° and 90° North.

1.1 Background

The Arctic and Subarctic are vast (Figure 1) and cover a variety of terrain and atmospheric conditions. The Army Technical Manual (TM) (DA 2017) introduces the general principles and technical guidance for performing field maintenance, operation, and repair of equipment under cold weather conditions specifically in below freezing temperatures. Also, the Army Regulation (AR) 70-38 (HQDA 2020) provides *Cold and Severe Cold Climatic Categories*. These areas include northern North America, Greenland, northern Asia, and Tibet. The categories for cold conditions in AR 70-38, are divided into four elements (basic cold, cold, severe cold, and extreme cold; Table 1) and represent only the upper and lower limits of the daily temperature cycles. The daily extremes represent the highest and

1. For a full list of the spelled-out forms of the units of measure used in this document, please refer to *US Government Publishing Office Style Manual*, 31st ed. (Washington, DC: US Government Publishing Office, 2016), 248–52, <https://www.govinfo.gov/content/pkg/GPO-STYLEMANUAL-2016/pdf/GPO-STYLEMANUAL-2016.pdf>.

lowest values in a 24 hour cycle of temperature, solar radiation, and relative humidity (RH). The AR 70-38 cold climatic category states,

The temperature during the coldest month in a normal year may be colder than the basic climatic category cold extreme of -32°C (-26°F). In the severe cold areas, the temperature during the coldest month in a normal year may be colder than the cold climatic category extreme of -46°C (-51°F). Temperatures colder than -51°C (-60°F) occur no more than 20 percent of the hours in the coldest month of the coldest part of the area (northern Siberia) where temperatures as low as -68°C (-90°F) have been recorded (HQDA 2020, 5).

Under conditions of low temperature and limited solar radiation, surfaces remain cold. Even with low to moderate absolute humidity, RH tends to reach saturation which promotes frost formation on exposed surfaces and equipment.

Figure 1. Arctic and Subarctic terrain characteristics.

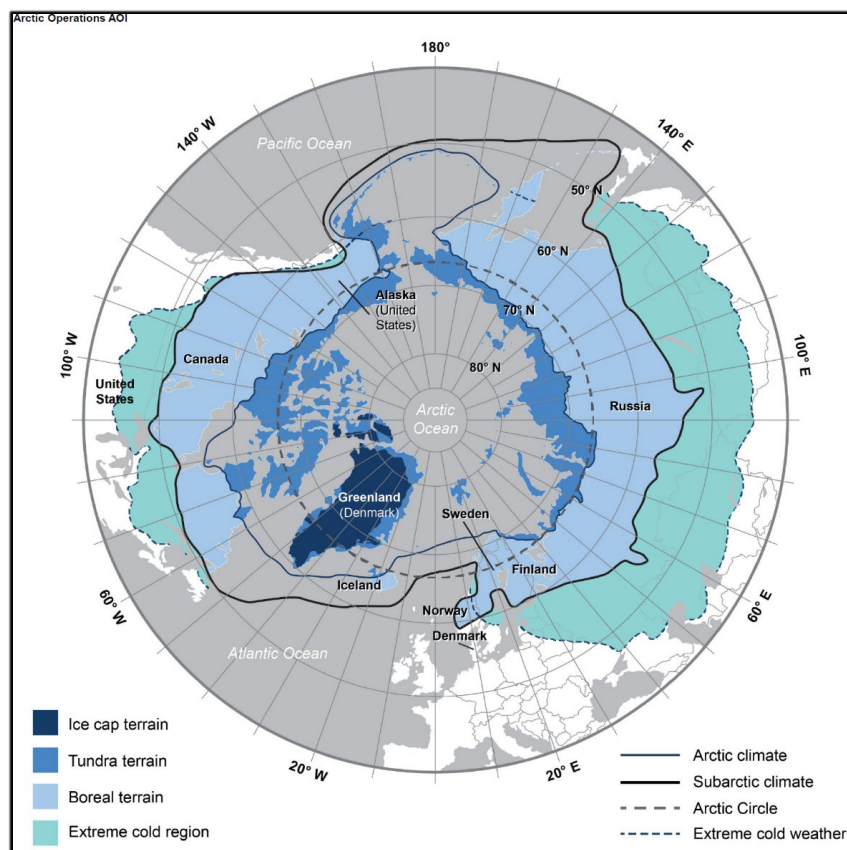


Table 1. Summary of cold condition limits as described in Army Regulation (AR) 70-38 (HQDA 2020).

Daily Cycle	Operational Conditions		Storage and Transit Conditions		Natural Environment Exposure Testing ^a °C (°F)
	Ambient Air Temperature, °C (°F)		Induced Air Temperature, °C (°F)		
	Daily Low	Daily High	Daily Low	Daily High	
Basic Cold (C1)	-32 (-25)	-21 (-5)	-33 (-28)	-25 (-13)	0 to -32 (32 to -25)
Cold (C2)	-46 (-50)	-37 (-35)	-46 (-50)	-37 (-35)	-32 to -46 (-26 to -50)
Severe Cold (C3)	-51 (-60)		-51 (-60)		<-46 (<-50)
Extreme Cold (C4)	-57 (-70)		-57 (-70)		

Note: C1—Operational design conditions are 5 continuous hours with an ambient air temperature of -25°F (-31°C), a minimum ground surface temperature of -35°F (-37°C), wind speed less than 16 ft/s (5 m/s). Design criteria for storage and transit are five continuous hours with an induced air temperature of -28°F (-33°C) with no wind or solar radiation and humidity tending toward saturation.

C2—Design conditions are 6 continuous hours with an ambient air temperature of -50°F (-46°C), a minimum snow surface temperature of -50°F (-46°C), wind speed less than 16 ft/s (5 m/s), negligible solar radiation (horizontal surface), and RH tending towards saturation.

C3—The design condition is a minimum temperature of -60°F (-51°C). (For testing purposes, this is a cold soak temperature.) Wind speeds are less than 16 ft/s (5 m/s).

Snow load design criterion—Portable equipment, such as tentage is based on 24 hour snowfalls withstanding a max snow load value of 10 lb/ft² (48.9 kg/m²), which is equivalent to a depth of 20 in. (508 mm) of snow with a specific gravity of 0.1. Temporary equipment such as rigid shelters, portable hangars, and so forth is based snowfalls associated with storms lasting longer than one day and is required to withstand a snow load value of 20 lb/ft² (97.7 kg/m²), which is equivalent to a snow depth of 40 in. (1,016 mm) with a specific gravity of 0.1. This equipment should be cleared of snow between storms. Semipermanent structure (dismountable) is based on seasonal accumulation of snow and is required to withstand a snow load value of 48 lb/ft² (235 kg/m²), which is equivalent to a snow depth of 96 in. (2,438 mm) with a specific gravity of 0.1.

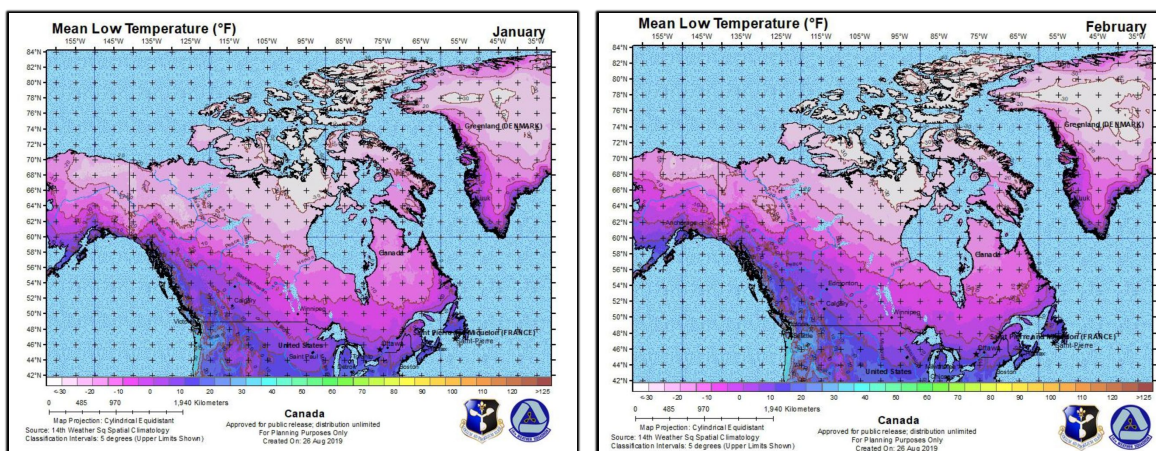
Wind—For operations, the following extremes, as given in MIL-HDBK-310 (DoD 1997) are a steady wind speed of 73 ft/s (22 m/s) and a gust of 95 ft/s (29 m/s).

^a Delineation to be used for natural environment exposure testing, not applicable to laboratory testing.

Moreover, the Army continues to identify stringent requirements for the Arctic operational environment, and expresses concerns of risk mission failure for sustaining the forces at temperatures reaching -65°F. This is because for some equipment and systems, the adverse effects of cold can occur even at temperatures above 32°F. However, the Arctic environment has wide variations in the state of the terrain and environmental conditions, and those conditions have vast divergence between the coastal areas and interior locations because of latitude, circulation patterns, and elevation. Modeled low mean temperatures of climatological data for January and February from the US Air Force, 14th Weather Squadron are shown in Figure 2. Using this, the Army requirement of operational temperatures reaching -65°F may only be encountered on rare occasions

in certain parts of the region. In addition to the temperature prerequisite, both the Army and the Air Force have rigid structural requirements for sustaining human habitation. Life sustaining structures must be able to withstand wind speeds of greater than 100 mph (125 mph at Pituffik Space Force Base, Greenland) and support 25 lb/ft² snow load. These snow load and wind levels may only be encountered in certain parts of the region. Therefore, focusing on operational requirements with unrealistic or impractical environmental parameters can preclude resources and capabilities from being applied wisely. The coincidence, or lack thereof, of these parameters will assist DoD Arctic operations planners with mission planning in order to apply the appropriate resources and logistics depending on reasonably expected atmospheric conditions.

Figure 2. Low mean temperatures for January (*left*) and February (*right*) from the US Air Force, 14th Weather Squadron. Obtained from the Climate Forecast System (CFS) v1, 1979–2010. (Images reproduced from USAF, n.d. Public domain.)



As a result, DoD planners continue to be challenged with these following overarching questions:

- *Are all Arctic material solutions required to operate at -65°F ?*
- *Are there existing capabilities and solutions that can operate at -65°F ?*
- *Are there existing capabilities that do not meet the -65°F requirement that can be used anywhere else in the Arctic?*
- *What regions of the Arctic and Subarctic never experienced +100 mph winds or 25 lb/ft² snow load?*

To answer the fundamental questions above, we developed characteristic northern latitude operational zones in the Arctic and Subarctic relative to these design thresholds.

1.2 Objective

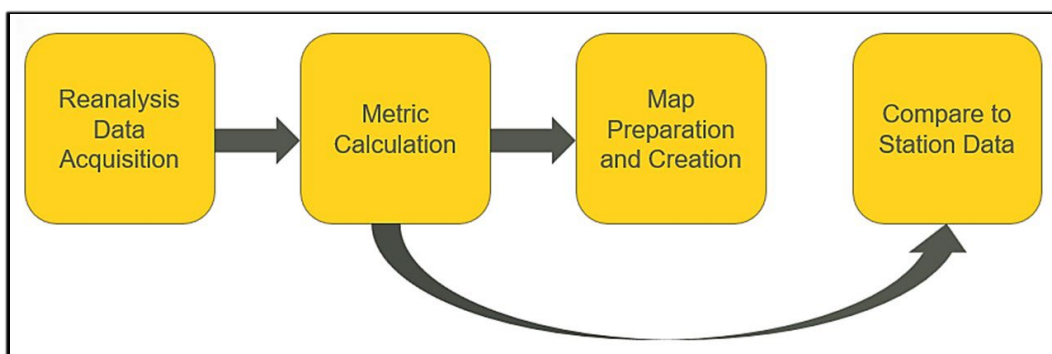
The objective of this effort is to further identify and refine the prevailing zonal characterization in the circumpolar north identified in TM 4-33.31 and AR 70-38. This zonal characterization includes three environmental parameters (air temperature extremes, snow depth, and wind speed limits) and mapped these against material and equipment exposure thresholds.

Operational zone characterization in the Arctic and Subarctic regions involves defining environmental limits to identify the contingency, logistics and resources needed, and to develop site-focused requirements for military operations in the Arctic and Subarctic regions. This provides mission-critical information that helps operations managers make risk-informed decisions based on the analysis of recent weather data.

1.3 Approach

To accomplish the intended project objectives, we used a 30-year span of meteorological data (described in the methodology below), validated and verified by observational point data. After the datasets are processed, we applied various metric calculations and generated maps for each parameter (Figure 3).

Figure 3. The steps used for Arctic zonal characterization.



The datasets used are described in Section . The processes or methodology for generating maps to visualize the environmental conditions for Arctic and Subarctic zones of operation is described in Section 2.3 as well as in Barrett et al (2025). Because cold environments are particularly difficult for military operations (soldiers, materials, equipment, etc.), this project also aimed to compile operational thresholding tables to map appropriate materiel to each zone of operation.

2 Methodology

2.1 Area of Study

The study region includes all areas between latitudes of 60°N and 90°N, rather than adhering to the established Arctic boundary at 66.5°N. Stakeholders requested this broader region to include both Arctic and Subarctic zones for analysis of a larger region of land with high mission activity, especially in the parts of Alaska and Canada between 60°N and 66.5°N latitude. As a result, the analyses and resulting statistics are not representative of the Arctic region alone or of any single nation, but rather the environmental patterns within the 60°N to 90°N band. This distinction is important for interpreting results and understanding the scope and applicability of the data.

2.2 Data

There are many datasets that can be used to determine atmospheric and terrain characterization parameters within the Arctic and near-Arctic. While point observations often provide the most accurate data at specific locations, data availability in the Arctic is typically both sparse and inconsistent because of extreme weather conditions, low population, financial considerations, and other factors. Military stakeholders, including mission planners at DoD installations, often have requirements to deploy in areas where there have been minimal meteorological observations. In these cases, the use of a numerical weather prediction (NWP) model analysis dataset with initialization values may be used.

Weather analysis datasets are an established and accepted procedure to project the state of the atmosphere from a finite set of imperfect, irregularly distributed observations onto a regular grid, or to represent the atmospheric state by the amplitude of standard mathematical functions, allowing for easy diagnostic study (AMS 2024a). An analysis may be considered as a space-time interpolation system. Essential ingredients of an analysis are a background field, usually a short-range forecast (although persistence may be an option), a forecast model, knowledge of the relative error of the many different observational platforms employed as well as knowledge of the error in the background field, and the spatial covariance of the various errors.

A specialized analysis known as reanalysis may provide an advantage over a standard analysis. Meteorological studies made with results from standard analyses may introduce inconsistencies from frequent changes in the NWP model configuration used to generate forecasts from previous analysis times. These changes (including changes in resolution and orography) lead to discontinuities in a time series of real-time analyses (AMS 2024b). A reanalysis produces complete, global gridded data that are as temporally homogeneous as possible. In the case of a reanalysis, the initial forecast (also known as the background field) is made by an NWP model that does not change over the entire period of the reanalysis. Reanalysis data may include many derived fields (heating, soil moisture over land, etc.) for which direct observations are nearly absent. Reanalysis combines model data with observations from across the world into a globally complete and consistent dataset using atmospheric physics. A reanalysis may produce data that goes several decades back, providing an accurate description of existing past atmospheric conditions.

2.2.1 European Center for Medium-Range Weather Forecasting (ECMWF) Reanalysis v5 (ERA5) as Baseline Gridded Dataset

According to AR 115-10 AFI 15-157 Interservice Publication (IP) Section 3-2, the United States Air Force (USAF) is statutorily responsible for meteorological operations in support of the Army Headquarters, Department of the Army (HQDA). This includes access to global surface and upper air meteorological observations, surface radar data, satellite imagery from geostationary, and polar orbiting sensors, as well as gridded datasets from NWP models. USAF has established multiple groups to assist with the variety of needs required for military applications. For this research, we relied on the expertise of the 14th Weather Squadron (14WS), located in Asheville, North Carolina. The 14WS is generally responsible for exploiting authoritative historical meteorological data to optimize military operations. After multiple conversations and the submission of a Support Analysis Request (SAR) (USAF 2017), 14WS recommended the usage of a reanalysis from the European Centre for Medium-Range Weather Forecasts (ECMWF) known as ECMWF Reanalysis v5 (ERA5) (Hersbach et al 2020).

ERA5 is a reanalysis dataset providing a consistent view of the evolution of meteorological and other parameters over several decades (ECMWF 2018). The reanalysis combines model data with observations from across the world into a globally complete and consistent dataset. ERA5 provides

hourly estimates for a large number of atmospheric and land surface parameters. A major strength of ERA5 is the much higher temporal and spatial resolution than those of previous global reanalyses.

The ERA5 dataset, as with any simulation, provides estimates which have some degree of uncertainty. Numerical models can only provide a representation of the real physical processes governing different components of the Earth System. In general, the uncertainty of model estimates grows larger as we go back in time, because the number of observations available to create a good quality atmospheric forcing is lower. The temporal resolution is hourly while the gridded spatial resolution is 0.25° (approximately 31 km). Thus, the ERA5 dataset provides an ideal, detailed snapshot of a variety of meteorological parameters.

2.2.2 Data Selection and Transformation

The parameters and threshold analyses was performed for the 60°N – 90°N latitude regions over a thirty year period. Data temporal range is based on the recognized international standard. The World Meteorological Organization (WMO) has established that thirty years is the standard time period required for calculating averages. Furthermore, WMO states that the most recent thirty-year period finishing in a year ending with 0 is the standard that should be utilized for data comparisons (WMO 2017). For this work, the years from 1991 to 2020 were chosen as the most recent available. This also serves to match other similar datasets produced by other meteorological agencies, such as the NOAA National Centers for Environmental Information, allowing additional dataset comparisons. It is expected that the products produced by this project would be updated on a decadal basis to continue to match future long term meteorological datasets. As the stakeholders for the project were primarily concerned with data and extremes occurring during winter months, only the months between October to March were considered for analysis.

The data subset used contained latitudes between 60°N to 90°N latitude to cover all Arctic regions plus portions of Subarctic regions. The data was obtained from the ECMWF Climate Data Store (also known as Copernicus) (ECMWF n.d.). The site provides free access to a variety of ECMWF maintained datasets; it allows both manual downloads and a Python API for more advanced data acquisition. The ERA5 data was initially obtained from ECMWF in gridded binary (GRIB) format. GRIB is a format based on standards established by the WMO and is typically used by operational

centers that produce NWP products. All parameters were downloaded in a single hourly GRIB formatted file. Parameters selected include 2 m height air temperature, skin (land surface) temperature, 10 m wind components, snow water equivalent (SWE), and level 1 soil temperature. Additionally, at the request of the stakeholders, snow depth was obtained from a derivative ECMWF dataset known as ERA5-Land (ECMWF 2025c). This dataset provides higher spatial resolution at 0.1° (approximately 9 km) at the expense of exclusion of data over oceanic and other water surface areas. Details about the collected variables are below:

- Skin temperature (SKT)—the temperature at the land surface, in Kelvin
- 2 m temperature (2T)—the temperature at a height of 2 m, in Kelvin
- Soil temperature, Level 1 (STL1)—the average temperature 0–7 cm below the surface, in Kelvin
- Wind velocity, *U*-Component (*U*)—the easterly wind velocity at a height of 10 m, in meters per second
- Wind velocity, *V*-Component (*V*)—the northerly wind velocity at a height of 10 m, in meters per second
- Snow depth (SD)—the depth of snow on the surface, in meters
- SWE—the amount of water stored in snow, in meters

Each GRIB file with multiple parameters was split into an individual parameter GRIB file using the open source wgrib software utility available from NOAA (NWS 2005). Upon parameter separation, each separated parameter GRIB file was then converted into Network Common Data Format (netCDF) format in an effort to increase compatibility with other data processing applications. The conversion was performed by the open-source National Center for Atmospheric Research (NCAR) Command Language (NCL) software (NCAR-NCL, n.d.). NCL allows users to perform a variety of statistical and visual analyses of meteorological datasets. The `ncl_convert2nc` tool included within the package allows for conversion of GRIB files to netCDF (NCAR-NCL 2019).

The wind speed (WSPD) and wind direction (WDIR) were calculated from the wind velocity components, using Equations (1) and (2), respectively (Earth Observing Laboratory, n.d.)

$$\text{WSPD} = \sqrt{V^2 + U^2}, \quad (1)$$

$$\text{WDIR} = \left(270^\circ - \arctan 2(V, U) \times (180^\circ) / \pi \right) \bmod 360^\circ. \quad (2)$$

The wind chill temperature (WCT), defined as the temperature felt to exposed human skin in Fahrenheit, because of the combined effect of air temperature (2T, in Fahrenheit) and wind speed (in miles per hour), was calculated using Equation (3) (Nelson et al. 2002),

$$\begin{aligned} \text{WCT} = & 35.74 + 0.6215(2T) - 35.75(\text{WSPD})^{0.16} \\ & + 0.4275(2T)(\text{WSPD})^{0.16}. \end{aligned} \quad (3)$$

The snow load (SL), defined as the downward force caused by the weight of accumulated snow or ice in pounds per square feet, was estimated using SWE (inches) assuming a horizontal angle with Equation (4) (Curtis, n.d.),

$$\text{SL} \approx \text{SWE} \times 5.2. \quad (4)$$

The frostbite time, defined as the time (in minutes) in which frostbite could occur on exposed, dry cheek skin at a given 2T (Fahrenheit) and 10 m wind speed (miles per hour) based on modeling, was calculated in addition to the meteorological variables using Equation (5) (Nelson et al. 2002),

$$\begin{aligned} \text{FT} = & \left(\left(-24.5 \left(\left(0.667 \left(\text{WSPD} \times \frac{8}{5} \right) \right) + 4.8 \right) \right) + 2111 \right) \\ & \left(-4.8 - \left((2T - 32) \times \frac{5}{9} \right)^{-1.668} \right). \end{aligned} \quad (5)$$

2.2.3 Point Observations Using Meteorological Aerodrome Report (METAR)

At the recommendation of project stakeholders, additional meteorological point observational data was obtained to provide limited verification of reanalysis data in key locations. These real-time observations are available in the International Civil Aviation Organization (ICAO) standardized Meteorological Aerodrome Report (METAR) format (ICAO 2021). Typically, each station reports meteorological parameters at a minimum of hourly thresholds, often through automated means. Thirty years of METAR observations were obtained for stations within the area of interest. In some cases, certain stations were not available for the entire period because of budgetary and technological considerations of the individual countries providing the observations.

Once the raw METAR observations were obtained, an initial processing of the data was undertaken utilizing the open-source python-metar software suite (Pollard 2018). Data was processed to extract temperature, dew point, visibility, wind speed and cardinal direction, and current meteorological phenomena observed at the time of the weather observation. The data was assembled in a comma separated value (CSV) format. Additionally, a date-time code was included which allowed METAR data to be more easily combined for the purposes of generating time series statistics.

2.3 Geospatial Analyses

The processes for geospatial analysis are described in detail by Barrett et al. (2025) SR-25, *Development and Management of Arctic Zonal Classification Products*. Below is a brief overview of the geospatial analyses performed. Section 4, “Geospatial Characterization Results” provides a snapshot of the results of the analysis, and example maps.

2.3.1 Metric Calculations

Four main metrics were calculated from the hourly ERA5 reanalysis data files across six different time scales to characterize prevailing conditions in the circumpolar north. The following time scales were used to calculate various statistics and metrics:

- Hourly
- Six-hour blocks (00:00–05:59 UTC, 06:00–11:59 UTC, 12:00–17:59 UTC, 18:00–23:59 UTC)
- Daily (00:00–23:59 UTC)
- Weekly (7–8 days)
- Biweekly (14–16 days)
- Monthly
- Seasonal (October–March)

The *average* metric represents the mean value of each meteorological parameter within a specific time period across all 30 years of data (January 1991–October 2020). For six-hour time periods (e.g., 3 December 00:00–05:59 UTC), the average was calculated using all hourly files within that period across the 30 year record (180 hourly files per six-hour period). For wind direction specifically, a circular mean was employed to account for the cyclical nature of directional data.

The *extreme* metric captures the single most severe value recorded within each time period across the entire 30 year dataset for each grid cell. For temperature variables (SKT, 2T, WCT, STL1), this represents the minimum recorded value. For WSPD, SD, and SL, the maximum value was extracted.

The *frequency* metric quantifies the percentage of single hours during which a variable exceeded a specified threshold over the course of the winter. The thresholds used are those specified in *Arctic Doctrine* Army Techniques Publications (ATP) 3-90.96, Table 1-3 and TM 4-33.31, Table 2, as well as the *Combat Weather Teams* stoplight chart. For each grid cell, the total number of hours exceeding the threshold was divided by the total number of hours in the time period and multiplied by 100 to express the result as a percentage.

The *length of consecutive extreme events* metric provides a detailed method for understanding how long intense environmental conditions persist. By analyzing thirty years of data, researchers track the duration of events that rise above a specific intensity threshold. For each day, the method identifies and measures separate periods when conditions exceed the threshold. First, the average duration of extreme events within each 24 hr period is calculated using the raw hourly reanalysis data. Then, these daily measurements are averaged across the entire 30-year dataset. The analysis excluded six-hour time scales because of insufficient temporal resolution. For time scales larger than daily (weekly, biweekly, monthly, and seasonal), the daily consecutive values were averaged across the respective period.

2.3.2 Generation of Static Maps

A standardized mapping workflow was implemented to ensure consistency across all visualizations. Custom map layout templates and symbology files were created to maintain visual coherence across all maps for each parameter, metric, and time scale combination. The maps were then evaluated and quality controlled manually (Barrett et al. 2025).

Further detailed information regarding the ERA5 reanalysis data variables and detailed process and methodology for map generation can be found in the US Army Engineer Research and Development Center (ERDC) Special Report, *Development and Management of Arctic Zonal Characterization Products* (Barrett et al. 2025).

3 Review of Thresholds

3.1 Zones and Thresholds

Another primary task was the compilation of operational thresholding tables to map which items, equipment, and materials can be used in each zone of operation, as defined in Table 2 below. The mapping shown previously identifies where in the Arctic certain environmental conditions are likely to be experienced at which parts of the winter season. Using these maps, the threshold for materiel and equipment detailed below gives mission planners information about which items can be operated (what their minimum temperature conditions are).

We followed and expanded the criteria for temperature ranges listed in the *Arctic Doctrine* ATP 3-90.96 (Table 2) (HQDA and USMC 2025) and TM 4-33.31 (Table 3) (DA 2017).

Table 2. Temperature zones. (Data from HQDA and USMC 2025; US Army 2025.)

Zone	Ranges	Descriptions from ATP 3-90.96 and TM 4-33.31
1	39 °F to 20 °F	Wet cold per ATP 3-90.96; a collection of cold weather clothing is necessary in layers to protect the Soldiers from the cold.
2	19 °F to -4 °F	Dry cold per ATP 3-90.96.
3	-5 °F to -24 °F	Intense cold per ATP 3-90.96; Soldier welfare considerations become an issue prompting equipment winterization when temperatures reach -10 °F. At -20 °F, metal becomes half as strong and can break, rubber stiffens and tires can form flat spots, and plastics become brittle and can shatter.
4	-25 °F to -40 °F	Extreme cold per ATP 3-90.96; almost all equipment undergoes testing for proper operation at temperatures reaching -25 °F. At temperatures below -30 °F, equipment begins to malfunction. At -40 °F exposed skin can freeze in under 10 min causing frostbite and impair equipment operations.
5A	-41 °F to -50 °F	At -50 °F tires become rigid enough to support a load at 0 pressure without apparent deflation; jet propulsion (JP) fuel (i.e., JP-8) under the Single Fuel Policy (SFP), is usable to -53 °F.
5B	-51 °F to -60 °F	When cooled gradually but continuously over a short time, rubber remains flexible until it reaches approximately -60 °F.
5C	Below -61 °F	A fully charged battery can withstand -70 °F to about -77 °F without freezing.

Table 3. Snow ranges for vehicle limitations. (Data from US Army and US Marine Corps 2025.)

Vehicle Type	Snow Depth (in.)
Wheeled	8
Wheeled 4×4 or greater	15
Wheeled 4×4 or greater with snow chains	20
Tracked	40
Over-snow	Over 40

In addition, there are human physiological challenges to operating in the Arctic, especially in dangerous windchill and very low temperatures that one would be susceptible to frostbite. Windchill is listed in ATP 3-90.96 (Table 4) and expanded the criteria for temperature ranges, particularly for temperature zone 5. ATP 3-90.96 describes zone 5 as hazardous conditions, indicating that the risks to military logistics and operations (including materiel, equipment, etc.) are significant. The most extreme temperature conditions are likely to be experienced in short durations and are shown in the results and the zonal statistics (see Barrett et al. 2025). Time of frostbite is also listed in ATP 3-90.96 (Table 5), and Equation (5) is applied to generate the maps for the Arctic (see Barrett et al. 2025). For wet skin, time to frostbite is much quicker than the time specified in Table 6 below.

Table 4. The wind chill temperature, in degrees Fahrenheit, from 10 m wind speed and 2 m air temperature. (Data from Nelson et al. 2002.)

–	–	Air Temperature (°F)																	
–	–	40	35	30	25	20	15	10	5	0	–5	–10	–15	–20	–25	–30	–40	–45	–50
Wind Speed (mph)	5	36	31	25	19	13	7	1	–5	–11	–16	–22	–28	–34	–40	–46	–52	–57	–63
	10	34	27	21	15	9	3	–4	–10	–16	–22	–28	–35	–41	–47	–53	–59	–66	–72
	15	32	25	19	13	6	6	–7	–13	–19	–26	–32	–39	–45	–51	–58	–64	–71	–77
	20	30	24	17	11	4	4	–9	–15	–22	–29	–35	–42	–48	–55	–61	–68	–74	–81
	25	29	23	16	9	3	3	–11	–17	–24	–31	–37	–44	–51	–58	–64	–71	–78	–84
	30	28	22	15	8	1	1	–12	–19	–26	–33	–39	–46	–53	–60	–67	–73	–80	–87
	35	28	21	14	7	0	0	–14	–21	–27	–34	–41	–48	–55	–62	–69	–76	–82	–89
	40	27	20	13	6	–1	–1	–15	–22	–29	–36	–43	–50	–57	–64	–71	–78	–84	–91
	45	26	19	12	5	–2	–2	–16	–23	–30	–37	–44	–51	–58	–65	–72	–79	–86	–93
	50	26	19	12	4	–3	–3	–17	–24	–31	–38	–45	–52	–60	–67	–74	–81	–88	–95
Frostbite Danger Level					Slight Danger (<i>green</i>)					Increased Danger (<i>orange</i>)					Great Danger (<i>red</i>)				
Time until potential frostbite in dry-exposed skin (min)					Less than 120					Less than 45					Less than 5				

Table 6. Stoplight chart that the Combat Weather Teams in Alaska use. (Data from personnel at Fort Wainwright, pers. comm., February 2026.)

Operation	Favorable (No Degradation)	Marginal (Some Degradation)	Unfavorable (Significant Degradation)
Airborne Operations Using Static Lines	Wind < 12 kn Ceiling (CIG) ≥ 2,000 ft No Precipitation Visibility (VIS) ≥ 4,800 m	CIG 1,000–1,999 ft Light–Medium Precipitation VIS 1,601–4,799 m	Wind > 13 kn CIG < 1,000 ft Heavy Precipitation VIS ≤ 1,600 m Thunderstorms (TSTMS)
Fixed Wing	Wind < 35 kn CIG > 2000 ft VIS > 3200 m	Wind 35–49 kn CIG 1,000–2000 ft VIS 1,600–3,200 m	Wind ≥ 50 kn CIG < 1,000 ft VIS < 1,600 m TSTMS
Rotary Wing	Wind < 30 kn CIG > 1,000 ft VIS > 4,800 m Light Turbulence (Turb)	Wind 30–44 kn CIG 700–1,000 ft VIS 1,601–4,800 m Medium Turb	Wind ≥ 45 kn CIG < 700 ft VIS ≤ 1,600 m TSTMS Medium–Severe Turb
Medevac Ops	Wind < 45 kn CIG ≥ 1,000 ft VIS ≥ 4,800 m	CIG 700–1,000 ft VIS 1,600–4,800 m	Wind ≥ 45 kn CIG < 700 ft VIS < 1,600 m
Gray Eagle (UAV)	Wind < 20 kn CIG > 10,000 ft No Precipitation VIS > 8,000 m Temp –14 °F to 109 °F Cross Wind (Xwind) < 9 kn Light Turb	Wind 20–25 kn CIG 5,000–9,999 ft Light Precipitation VIS 4,800–8,000 m Temp –15 °F to –24 °F Temp 109 °F to 119 °F Xwind 9–26 kn Medium Turb	Wind ≥ 26 kn CIG < 5,000 ft Medium–Heavy Precipitation VIS < 4,800 m TSTMS Temp ≤ –25 °F Temp ≥ 120 °F Xwind > 26 kn Severe Turb

Table 6 (cont.). Stoplight chart that the Combat Weather Teams in Alaska use. (Data from personnel at Fort Wainwright, pers. comm., February 2026.)

Operation	Favorable (No Degradation)	Marginal (Some Degradation)	Unfavorable (Significant Degradation)
Shadow Uncrewed Aerial Vehicle (UAV)	Wind < 15 kn CIG > 6,000 ft No Precipitation VIS > 4,800 m Temp < 105 °F	Wind 15–25 kn CIG 5,000–6,000 ft Light Precipitation VIS 1,601–4,800 m Temp 105 °F to 119 °F	Wind > 25 kn CIG < 5,000 ft Medium– ≤ 1,600 m TSTMS Temp Heavy Precipitation VIS ≥ 120 °F
Personnel	No Precipitation Temp 20 °F to 85 °F	Light Precipitation Temp –15 °F to 19 °F Temp 85 °F to 95 °F	Medium–Heavy Precipitation Temp < –15 °F Temp > 95 °F TSTMS
Air Trafficability (flight level <1,000 ft)	No Precipitation < Light Icing VIS > 4,800 m	Light Precipitation Light–Medium Icing VIS 1,601–4,800 m	Medium–Heavy Precipitation Severe Icing VIS ≤ 1,600 m
Air Assault	Wind < 25 kn CIG ≥ 1,000 ft VIS ≥ 5,000 m	Wind 25–40 kn CIG 300–1,000 ft VIS 800–5,000 m	Wind ≥ 40 kn CIG < 300 ft VIS < 800 m
CH47 Sling Load	Wind ≤ 14 kn	Wind 15–30 kn Light Turb	Wind > 30 kn Light Turb
CH60 Sling Load	Wind ≤ 8 kn	Wind 9–14 kn Light Turb	Wind > 15 kn Light Turb
Forward Arming and Refueling Point (FARP) Ops	Wind ≤ 30 kn	Wind 30–44 kn	Wind ≥ 45 kn
Illumination	—	Lunar Illumination ≥ 25% And Lunar Elevation 0°–30°	Lunar Illumination < 25% And Lunar Elevation < 0°

3.2 Arctic Military Equipment and Gear for Specific Zones

Operations in extreme Arctic environments require unique clothing and equipment solutions specific to each of the defined areas and variations in terrain. For example, military operations in the boreal forests of Scandinavia and elsewhere, rely heavily on skis and snowshoes and tracked vehicles to facilitate maneuver in deep snow conditions. Soldier clothing must be light-weight, well insulated, and suited to wet snow conditions. Camouflage patterns must be matched to snow covered forests with shades of white, green, and brown. Soldiers are able to operate using unheated shelters for short periods of time and overhead tree cover helps to mask Soldier movements and locations.

Operations in the extreme cold of the Arctic tundra and on sea ice requires more heavily insulated clothing and camouflage patterns matched to the whites, blues, and grays of a snowy environment devoid of trees. Movement is largely restricted to snowmobiles and other tracked vehicles, and Soldiers will require heated shelters for all but the shortest durations.

Soldiers that operate in cold environments, such as the 11th Airborne Division (ABN DIV) stationed in Alaska are issued specialized winter uniforms, allowing them to operate in extreme environments. Soldiers are issued a seven-layer clothing system called the Extended Cold Weather Clothing (ECWC) System. This system is adaptable and can be worn in multiple configurations to ensure proper insulation and decrease the likeliness of cold weather injuries.

Level 1 is a lightweight, long sleeve top and full-length bottoms, constructed out of a silk-weight moisture-wicking polyester, they are commonly referred to as *silks*. Level 2 is a mid-weight insulation layer made of a polyester *grid* fleece design. Level 3 is a fleece cold weather jacket that can be used as either the primary insulative layer or an outer garment. Level 4 is a wind cold weather jacket made of a lightweight, wind resistant, and water-repellant material. This is a minimum outer shell layer. Level 5 is a soft-shell outer layer jacket and trousers. Level 6 is the extreme cold or wet weather jacket and trousers. These are waterproof and used in prolonged or hard rain, typically used when temperatures are alternating between freezing and thawing. Level 7 is the extreme cold weather parka and trousers. These are the warmest of the outer layers and

are used to provide superior warmth while still being packable and lightweight (TM 10-845-236-10; DA 2014).

According to Army TM 10-845-236-10, the ECWC shall be worn in the following configurations. Table 7 shows the configurations for active wear of the system and Table 8 shows the configurations for static wear.

Table 7. Active wear of the Extended Cold Weather Clothing (ECWC) System. (Data from Technical Manual [TM] 10-8415-236-10; DA 2014.)

Temperature	Zone	Level 1	Level 2	Level 3	Level 4	Level 5	Level 6	Level 7
Cold 0°F to 30°F	1,2	X	X	—	—	X	—	—
Wet 40°F ± 0°F	1	X	—	—	X	X(B)	—	—
Cold/Wet 35°F to 45°F	1	X	—	—	—	X	—	—
Extreme Cold -25°F to 0°F	3	X	X	X	—	X	—	—

Note: (T)—Top; (B)—Bottom; X—Both top and bottom worn.

Table 8. Static wear of the ECWC System. (Data from TM 10-8415-236-10; DA 2014.)

Temperature	Zone	Level 1	Level 2	Level 3	Level 4	Level 5	Level 6	Level 7
Cold/Wet 30°F to 45°F	1	X	—	X	—	—	—	X
Wet above 45°F	1	X	X(T)	—	—	—	X	—
Cold 0°F to 30°F	1,2	—	X	—	—	X	—	X(T)
Extra Cold -50°F to 0°F	2,3,4,5ABC	X	X	X	—	X	—	X

Note: (T)—Top; (B)—Bottom; X—Both top and bottom worn.

There are other worn items and gear that are not covered in TM 10-845-236-10 such as gloves, boots, and sleeping systems. Temperatures thresholds for these items (Table 9) were gathered from subject matter experts at the Army's Central Issue Facility (CIF) on Fort Wainwright, Alaska, and are based on what is currently issued to Soldiers stationed there.²

2. T. T. Richie, pers. comm., 26 March 2025.

Table 9. Operational temperature limits of worn gear.

Gear	Max Operation Zone	Minimum Temperature
Standard Issue Light Duty Leather gloves	Zone 1	35 °F
Arctic Mittens (with Liners)	—	Unknown ^a
OR Mittens (with liners)	—	Unknown ^a
OR Trigger Finger Mittens	—	Unknown ^a
OR Convoy Gloves	—	Unknown ^a
OR Contact Gloves	—	Unknown ^a
Vapor Barrier Boots **	Zone 5	–50 °F
400 g Danner Boots ^b	Zone 3	–20 °F
600 g Danner Boots ^b	Zone 3	–20 °F
1200 g Danner Boots ^b	Zone 3	–20 °F
Mukluk Boots ^c	Zone 4 or 5C	–40 °F or –100 °F
Type I: Temperate Sleeping Bag Inner	Zone 2	≥15 °F
Type II: Cold, Sleeping Bag: Stand Alone	Zone 2	5 °F to 20 °F
Type III: Type II Used Over Type I	Zone 3	–13 °F to 5 °F
Modular Sleeping Bag Intermediate Cold	Zone 2	0 °F to 32 °F
Sleep System: Complete Wearing ECWCS	Zone 5A	–50 °F

Note: Zones 1–5C described in Table X above (3.1, Zones and Thresholds).

^a Companies will not commit to temperature thresholds for legal reasons.

^b All boot ratings are assuming Soldiers are wearing Darn Tough Boot Heavyweight Tactical Sock, which is 80% merino wool, 19% nylon, and 1% Lycra spandex.

^c Mukluk temperature depends on rating of the inserts inside the boot.

It is important to recognize that these thresholds are general guidelines, and actual minimum tolerance levels can vary significantly between individuals. Each Soldier has a unique response to low temperatures from personal comfort, physiology, acclimation, and experience. Caution is advised when operating near the minimum effective limits of issued gear. It is always best to avoid excessive sweating and moisture buildup by maintaining a state of comfortably cool/warm—cool enough to prevent overheating, but warm enough to sustain performance and safety.

Each piece of equipment has a different temperature threshold (Table 10) depending on the chemical and physical properties. The following tables (Tables 10–30) outline different equipment and solutions that is readily used in units across the military and their minimum operational thresholds according to Army TM 4-33.31, *Cold Weather Maintenance Operations* (DA 2017).

Table 10. Equipment. (Data from TM 4-33.31; DA 2017.)

Equipment	Max Operation Zone	Minimum Operational Temperature
Electrical and Ignition Systems	Zone 4	−30 °F
Vehicle Crew Heaters	Zone 4	−25 °F
Personnel Heaters	Zone 4	−40 °F
Level Vials	Zone 4	−40 °F
Cold Soaked Vehicles	Zone 4	−25 °F

Cold-soaked vehicles refer to any vehicle that is left in the elements, without being turned on and operated, for an extended period of time, such as 24 hours or more. When operating in temperatures lower than −25°F, it is imperative to warm any cold soaked vehicle prior to attempting to start or operate the vehicle. Although the winterization kit is not designed to start a vehicle engine after prolonged exposure to extreme temperatures, it can aid in getting an engine started if adequate time and auxiliary power are available to heat critical vehicle components (TM 4-33.31, HQDA 2017).

Most tents and shelters used by the military are manufactured by civilian organizations and each tent will have different operational thresholds. Table 11 outlines some of the tents used by the 11th ABN DIV, these tents are all from *HDT Global* and thresholds are according to their sales representatives.³

Table 11. Tents.

Tent	Max Operation Zone	Minimum Operational Temperature
Arctic 10-Man Tent (Ahkio)	Zone 5A	−40 °F
HDT Airbeam	Zone 4	−25 °F
HDT Airbeam (Arctic)	Zone 5C	−65 °F
Base-X	Zone 4	−40 °F
Deployable Rapid Assembly Shelter (DRASH)	Zone4	−40 °F

Note: Zones 1–5C described in Table 10 above (3.1, Zones and Thresholds).

Batteries are integral to most modern military equipment, and their performance can be significantly impacted by low temperatures. To ensure optimal functionality, proper cold weather charging procedures must be followed. Batteries should be fully charged when operating in extreme cold as low temperatures can severely reduce capacity and efficiency. Prolonged exposure to freezing conditions may cause batteries to freeze posing a risk

3. R. Zamora, pers. comm., 2 May 2025.

of damage or even explosion if used prematurely. If freezing is suspected, batteries should be gradually warmed above 32°F before use. Always refer to manufacturer guidelines for specific operational thresholds. Table 12 lists several common military battery types along with their minimum operation thresholds.

Table 12. Batteries. (Data from TM 4-33.31; DA 2017.)

Battery	Max Operation Zone	Minimum Operational Temperature
Battery Storage	Zone 2	10 °F
Absorbed Glass Mat (AGM): Fully Charged	Zone 5C	-70 °F
AGM: Not Fully Charged	Zone 2	10 °F
Fully Charged Battery	Zone 5C	-70 °F to -77 °F
Gel Batteries: Fully Charged	Zone 5C	-70 °F
Lithium Battery (Li-SO ₂)	Zone 5B	-58 °F
Lithium Battery (LiMnO ₂)	Zone 3	-22 °F
Rechargeable Dry Cell Batteries Type I and Type II	Zone 2	-4 °F
Rechargeable Dry Cell batteries Type III and Type IV	Zone 1	-22 °F
Wet Cell Batteries: Fully Charged	Zone 5C	-75 °F

Electronic devices are ubiquitous across all branches of the military, playing a critical role in operations. However, their performance often degrades in cold environments. Table 13 highlights several commonly used Army electronics and their associated low-temperature limitations.

Table 13. Electronics. (Data from TM 4-33.31; DA 2017.)

Electrical Equipment	Max Operation Zone	Minimum Operational Temperature
Larger Workstation Computers	Zone 2	0 °F
Handheld computers	Zone 4	-30 °F
Ruggedized military versions of display or projection	Zone 2	-4 °F
LCD Screens	—	Check Manufacturer Specs
Digital Light Processing Projectors	Zone 4	-40 °F
Mortar Ballistic Computer	Zone 5A	-50 °F
RG-8 Cable	Zone 3	-20 °F
RG213 Coaxial Cable	Zone 4	-40 °F

Generators provide critical power for military operations in harsh environments. In Arctic conditions, there are multiple factors that can affect the performance of military generators such as, fuel viscosity, battery cranking power, and oil gelling. To ensure reliable cold weather operation, generators should be prewarmed when possible, and winterization kits should be installed and used when available. Table 14 outlines minimum temperature thresholds for many common military generators.

Table 14. Generators. (Data from TM 4-33.31; DA 2017.)

Generator	Max Operation Zone	Minimum Operational Temperature
Mobile Electric Power (MEP): Nonwinterized	Zone 4	-25 °F
MEP: Winterized	Zone 5A	-50 °F
Generator Storage	Zone 5B	as low as -60 °F
Military Tactical Generator (MTG)	Zone 5A	-50 °F
Tactical Quiet Generator (TQG): Nonwinterized	Zone 4	-25 °F
TQG: Winterized	Zone 5A	-50 °F
Advanced Medium Mobile Power Sources (AMMPS)	Zone 5A	-50 °F
Power Distribution and Illumination System, Electrical	Zone 4	-25 °F

To maintain the effective operation of military equipment in cold environments, it is essential to use the appropriate fuels, oils, lubricants, greases, and solvents. Each of these fluids has specific temperature thresholds that must be observed to ensure reliable engine performance and prevent damage. Tables 15–21 provide detailed information on the cold weather operational limits for commonly used fluids.

Table 15. Fuels. (Data from TM 4-33.31; DA 2017.)

Fuel	Max Operation Zone	Minimum Operational Temperature
Jet Propulsion Fuel (JP-8)	Zone 5B	-53 °F
F-24	Zone 4	-40 °F
Jet A	Zone 4	-40 °F
Jet A-1	Zone 5B	-52.6 °F
Jet B	Zone 5B	-58 °F
JP-5	Zone 5B	-51 °F
JP-4	Zone 5C	-74 °F
Antarctic Jet Fuel (AN-8)	Zone 5C	-74 °F

Table 16. Engine oils, by military symbol. (Data from TM 4-33.31; DA 2017).

Engine Oil (Military Symbol)	Max Operation Zone	Minimum Operational Temperature
OE/HDO-SCPL	Zone 5B	-58 °F
OEA-30	Zone 5B	-58 °F
OE/HDO 40	Zone 1	20 °F
OE/HDO-10/30	Zone 3	-15 °F
OE/HDO-5/40	Zone 4	-30 °F
OE/HDO-15/40	Zone 2	0 °F
15 W 40 (Transmission Oil)	Zone 3	-10 °F

Table 17. Hydraulic fluids listed by standard name then military symbol (if applicable). (Data from TM 4-33.31; DA 2017.)

Hydraulic Fluids (Standard / Military Symbol)	Max Operation Zone	Minimum Operational Temperature
MIL-PRF-6083 / OHT	Zone 5C	-65 °F
MIL-PRF-46170 / FRH	Zone 4	-40 °F
MIL-PRF-5606 / OHA	Zone 5C	-65 °F
MIL-PRF-83282	Zone 4	-40 °F
MIL-PRF-87257	Zone 5C	-65 °F
MIL-PRF-46176	Zone 5C	-67 °F

Table 18. Gear lubricating oils, by military symbol. (Data from TM 4-33.31; DA 2017.)

Gear Lubricating Oils (Military Symbol)	Max Operation Zone	Minimum Operational Temperature
G0-75	Zone 1	50 °F
G0-80/90	Zone 4	-18 °F
G0-85/140	Zone 2	4 °F
G0-75	Zone 5A	-50 °F

Table 19. Grease lubricants. (Data from TM 4-33.31; DA 2017.)

Grease Lubricants	Max Operation Zone	Minimum Operational Temperature
Grease, Automotive, and Artillery	Zone 5C	−65 °F
Grease, Aircraft, General Purpose, Wide Temperature Range (Grease)	Zone 5C	−65 °F
Grease, Aircraft and Instrument, Gear, and Actuator Screw	Zone 5C	−99 °F
Grease, Molybdenum Disulfide, for Low and High Temperatures	Zone 5C	−100 °F
Grease, Graphite	Zone 3	−9 °F

Table 20. Solvents and cleaners, listed by military symbol. (Data from TM 4-33.31; DA 2017.)

Solvents and Cleaners (Military Symbol)	Max Operation Zone	Minimum Operational Temperature
Rifle Bore Cleaner (RBC)	Zone 1	32 °F
Type II High Flash Point Degreasing Solvent (SD-2)	—	n/a

Note: n/a—not applicable.

Table 21. Miscellaneous liquids. (Data from TM 4-33.31; DA 2017.)

Liquid	Max Operation Zone	Minimum Operational Temperature
Propylene-Glycol Deicing	Zone 5B	−60 °F
Propane	Zone 4	−40 °F
Arctic Antifreeze	Zone 5C	−90 °F
Wheel Bearing Lubricants	Zone 5C	−65 °F
Lubricating Oil, Internal Combustion Engine, Arctic	Zone 5C	−67 °F
Ethylene-Glycol-Distilled Water Mixture	Zone 5A	−50 °F to 32 °F
Arctic Antifreeze (Ethylene-Glycol Based)	Zone 5C	−40 °F to −90 °F

Note: For all petroleum, oils, and lubricants (POL), DoD does testing to −65 °F only.

Weapon systems are designed to function at all temperatures. In Arctic environments, it is imperative that any weapons remain outside of warmed tents. This helps prevent condensation from forming on and in these weapons, impeding their functionality. When weapons are left outside, they should be covered and sheltered from possible moisture. Cold weather affects the trajectory of rounds, impacting the maximum

effective range of each weapon system differently. Table 22 shows some thresholds for various weapon system components.

While there is not a specified minimum operational temperature for all weapons systems, these thresholds are based on normal Arctic operations and field experience. It is recommended to consult the appropriate technical manuals that outline specifics for each weapon system.

Table 22. Weapon systems and components. (Data from TM 4-33.31; DA 2017.)

Weapon System	Max Operation Zone	Minimum Operational Temperature	Technical Manual/Source
Level Vials	Zone 4	-40 °F	TM 4-33.31
Machine Guns	Zone 4	-40 °F	TM 9-1005-313-10 and TM 9-1005-201-10
Rifles	Zone 4	-40 °F	TM 9-1005-319-10
Rockets	Zone 4	-40 °F	TM 4-33.31

Each weapon system has its own lubrication requirements and requires different lubricants at different temperatures. Lubricants themselves have temperature thresholds (Table 23). The temperatures in this table are the lower limitation temperature.

Table 23. Weapons lubricants. (Data from TM 4-33.31; DA 2017.)

Weapons Lubricants (Military Symbol)	Max Operation Zone	Minimum Operational Temperature
PL-S	Zone 5A	-49 °F
Cleaner Lubricant Preservative (CLP)	Zone 5B	-60 °F
Lubricant, Small Arms (LSA)	Zone 5C	-65 °F
Lubricant, Weapon Semifluid (High-Load-Carrying Capacity) (LSAT)	Zone 4	-30 °F
Lubricant Arctic Weight (LAW)	Zone 5C	-70 °F

Rifle lubrication changes at different intervals. It is imperative to use proper lubricants to avoid gelling and burning which can occur if lubricants are used at improper temperatures (see Table 24). When switching from one lubricant to another, ensure the weapon is void of all other lubricants to avoid cross contamination which may cause weapons malfunctions.

Table 24. Rifle Lubrication guidelines. (Data from TM 4-33.31; DA 2017.)

Lubricant	Max Operation Zone	Temperature Range
CLP	Zone 3	$>-10^{\circ}\text{F}$
LSA	Zone 3	$>-10^{\circ}\text{F}$
LAW	Zone 1	$\leq 10^{\circ}\text{F}$

The M249, squad automatic weapon (SAW) should be lubricated with cleaner lubricant preservative (CLP) and is effective down to -30°F (Table 25). At lower temperatures, the weapon system is more prone to malfunctions and jams.

Table 25. The M249, squad automatic weapon (SAW) lubrication guidelines. (Data from TM 4-33.31; DA 2017.)

Lubricant	Temperature Range
CLP	$\geq -30^{\circ}\text{F}$

Machine guns, such as the M2 and M240B should be lubricated as outlined in Table 26. Generally, lubricant, weapon semifluid (high-load-carrying capacity) (LSAT) and grease, molybdenum disulfide (GMD) are interchangeable for machine guns.

Table 26. Machine gun lubrication guidelines. (Data from TM 4-33.31; DA 2017).

Lubricant	Temperature Range
LSAT	$\geq -20^{\circ}\text{F}$
Grease, Molybdenum Disulfide, for low and high temperatures (GMD)	$\geq -20^{\circ}\text{F}$
LAW	$\geq -20^{\circ}\text{F}$

To ensure effective range of mortars, they should be lubricated as outlined in Table 27. Failure to lubricate them properly can result in misfires, and decreased accuracy of the weapon system, posing a significant risk to friendly troops who may be operating in the vicinity of the impact zone.

Table 27. Mortar lubrication guidelines. (Data from TM 4-33.31; DA 2017.)

Lubricant	Temperature Range
General Purpose Lubricant (GLP)	>10 °F
LAW	<10 °F
Preservative Lubricant (PL) Special Purpose (SP)	<0 °F

Artillery pieces should be lubricated with CLP with temperatures reaching -10°F (Table 28). At low temperatures the maximum effective range of the weapon system is degraded and the M119 and the M777 are affected differently. Since they each have different systems, certain precautions are necessary for winterizing the components of each gun. Follow guidance in lubrication orders (LOs) and TMs for each system.

Table 28. Artillery lubrication guidelines. (Data from TM 4-33.31; DA 2017.)

Lubricant	Temperature Range
CLP	$\geq -10^{\circ}\text{F}$

For rotary-wing aircraft, operating limits are based on Soldier experience and are assembled in Table 29. Aircraft have been successfully started and flown in temperatures as low as -40°F (-40°C). That said, factors such as rotor wash and high-altitude wind chill can significantly lower cockpit and cabin temperatures, further complicating operations when cumbersome extreme cold weather gear must be worn. In such conditions, mission execution is subject to the commander's risk tolerance, the criticality of the mission, and the safety of aircrew and support personnel. In extreme cold scenarios, multiple environmental and mechanical factors can increase risk, requiring informed, situationally driven decisions by both command and ground personnel.

Table 29. Helicopters, according to Soldier experience.

Helicopter	Max Operation Zone	Minimum Operational Temperature
CH-47 Chinook	Zone 4	-40°F (-40°C)
AH-64 Apache	Zone 4	-40°F (-40°C)
UH-60 Blackhawk	Zone 4	-40°F (-40°C)

Note: The true limits have not yet been determined. Operational thresholds are limited to commander risk level and Soldier safety. Extreme cold poses an exponential risk to Soldiers inside of aircraft.

Fixed wing aircraft play a vital role in military operations, offering rapid transport, reconnaissance, and logistical support. The Air Force has been flying large cargo aircraft on successful missions to McMurdo Station, South Pole, for years. However, in these cold environments their performance and reliability can be significantly impacted. Successful operations require meticulous preflight planning, proper cold weather maintenance, and selection and use of the correct fuel for the temperatures. Table 30 outlines the known minimum operational thresholds for each group of aircraft. Knowing and understanding these operational limits will ensure crew safety and mission success. For most aircraft, the majority of minimum operational testing is on the electrical components within the cockpit. However, overall temperature related limitations in aircraft are largely driven by the properties and performance of petroleum, oil, and lubricants in the system.

Refer to Air Force Manual (AFMAN) 11-202V3 for temperature correction charts and other pertinent information (DAF 2022). It is also strongly recommended to consult the applicable AFMAN for each specific aircraft variant prior to conducting any flight planning in Arctic conditions. These manuals provide critical guidance on operational limitations, cold weather procedures, and safety considerations essential for mission success in extreme environments. Some specific guidelines are as follows:

- Do not take off with ice, snow, frost, or other contamination adhering to wings, control surfaces, propellers, engine inlets etc. (DAF 2022).
- Take off under freezing rain should be avoided (DAF 2024).
- For C-17 planning purposes flight managers shall not plan any flights to take place for known temperatures of -76°F (DAF 2024).

Table 30. Fixed wing aircraft.

Aircraft	Max Operation Zone	Minimum Operational Temperature
LC-130	Zone 5C	-65°F
C-17	Zone 5B	-76°F

Any cold-soaked aircraft below -25°F should be warmed with installed winterization kit before starting the aircraft.

4 Geospatial Characterization Results

4.1 Temperature

4.1.1 Skin Temperature (SKT)

The average SKT across the Arctic and Subarctic land mass between October and March was -9.0°F according to the 1991–2020 hourly measurements. The lowest temperatures were predominately in northern Greenland and Canada as well as Siberia (Figure 4). The average surface temperature in Scandinavia, western Russia, southern Canadian Arctic and Subarctic, and southern Alaska did not get below -4°F within the band of 60°N and 90°N (Table 31). February was the coldest month based on these average temperatures. The average lowest recorded SKT across the entire 60°N to 90°N landmass between October and March was -62.6°F from the 1991–2020 hourly measurements (Figure 5). Below -60°F surface temperatures occurred across the Arctic and Subarctic countries except for Scandinavia, which had a low between -50°F and -60°F , and Iceland, which did not experience temperatures below -40°F (Figure 5; Table 31). February and January had the lowest extreme temperatures (Figure 5).

Table 31. Zonal statistics per location (within 60°N – 90°N) for the average surface temperature, defined as the six-month mean (October–March), using hourly data from 1991–2020.

Location	Min ($^{\circ}\text{F}$)	5th PCT ($^{\circ}\text{F}$)	Mean ($^{\circ}\text{F}$)	95th PCT ($^{\circ}\text{F}$)	Max ($^{\circ}\text{F}$)
Alaska	-10.7	-7.0	4.9	19.9	39.3
Russia	-29.1	-19.7	-4.7	20.2	30.3
Finland	16.2	18.2	23.7	29.9	35.7
Sweden	15.3	17.2	23.5	30.9	34.7
Norway	16.0	18.0	25.1	35.0	41.4
Svalbard	2.3	3.2	8.9	17.1	29.0
Iceland	17.9	23.5	28.8	34.4	39.0
Greenland	-33.4	-31.9	-16.4	7.9	27.3
Canada	-28.5	-20.2	-8.5	6.5	16.2

Note: PCT = percentile.

The average lowest recorded SKT across the entire 60°N to 90°N landmass between October and March was -62.6°F from the 1991–2020 hourly measurements (Figure 5). Below -60°F surface temperatures occurred across the Arctic and Subarctic countries except for Scandinavia, which had a low between -50°F and -60°F , and Iceland, which did not experience temperatures below -40°F (Figure 5; Table 32). February and January had the lowest extreme temperatures (Figure 5).

The average frequency of surface temperatures being below -40°F was 10.57%, with the highest frequency occurring in central Greenland, northern Canada, and northern Greenland (Figure 6). These temperatures occurred most frequently in February, where parts of northern Greenland had frequencies of 92% (Figure 6).

Figure 4. Monthly average skin temperature (SKT) between 60°N and 90°N for October (A), November (B), December (C), January (D), February (E), and March (F), from hourly data between 1991 and 2020.

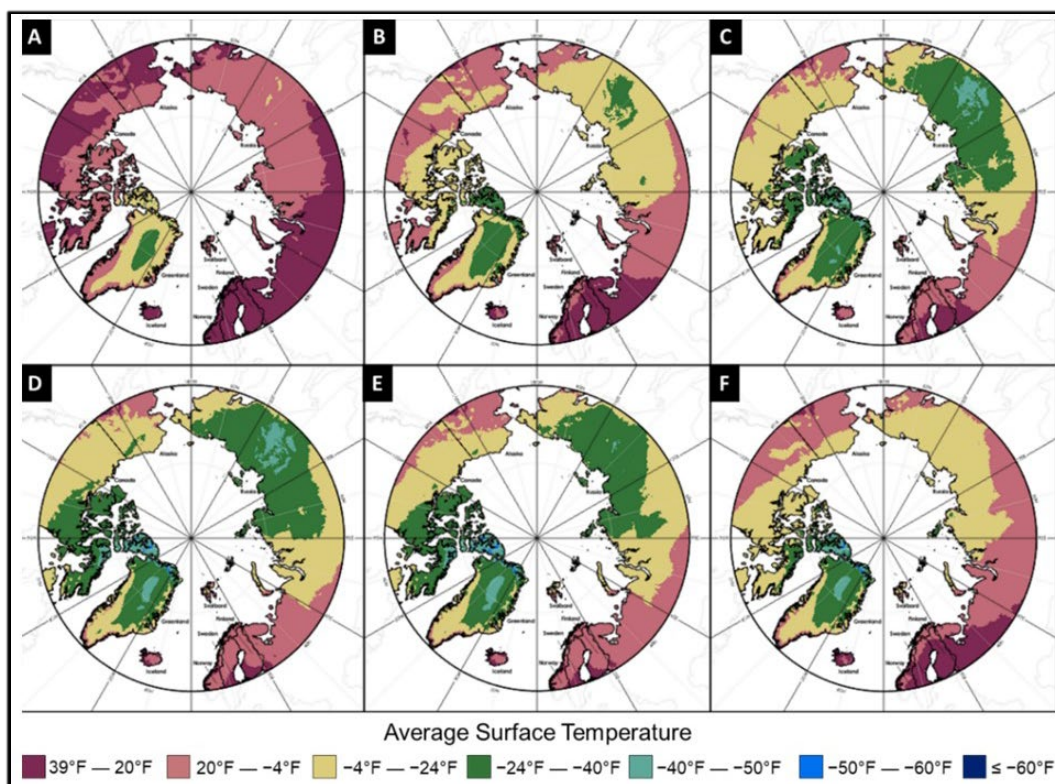


Figure 5. Monthly lowest reanalysis hourly SKT between 60°N and 90°N latitude for October (A), November (B), December (C), January (D), February (E), and March (F), from hourly data between 1991 and 2020.

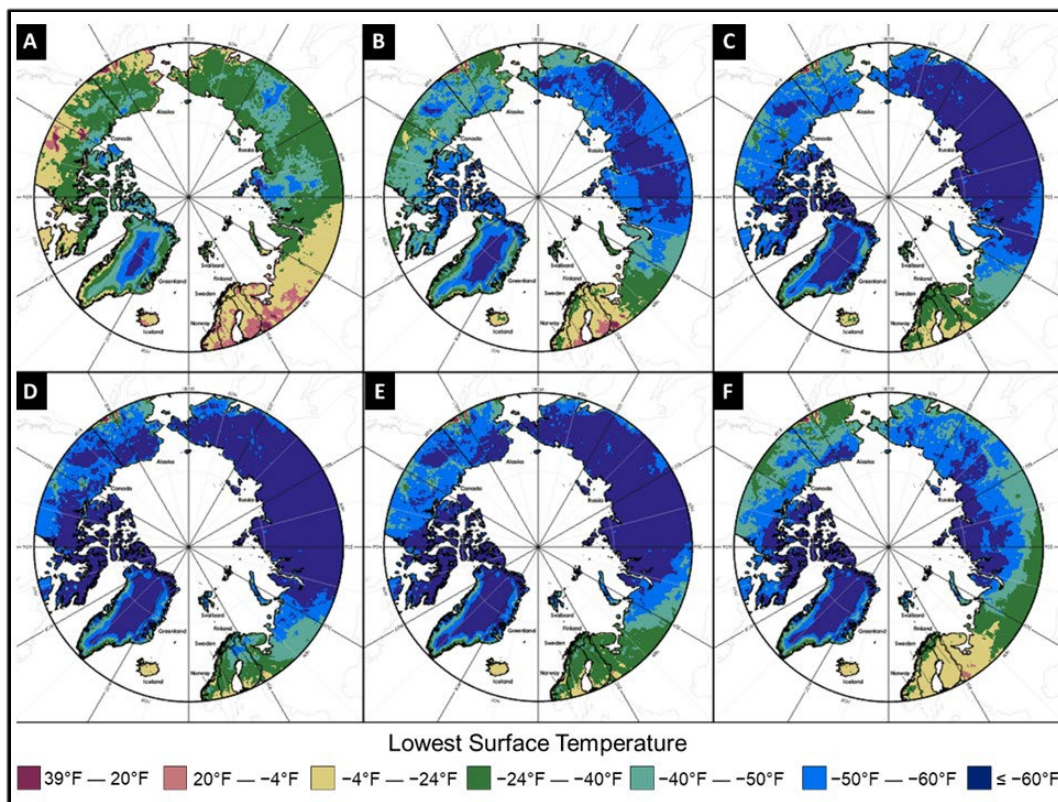
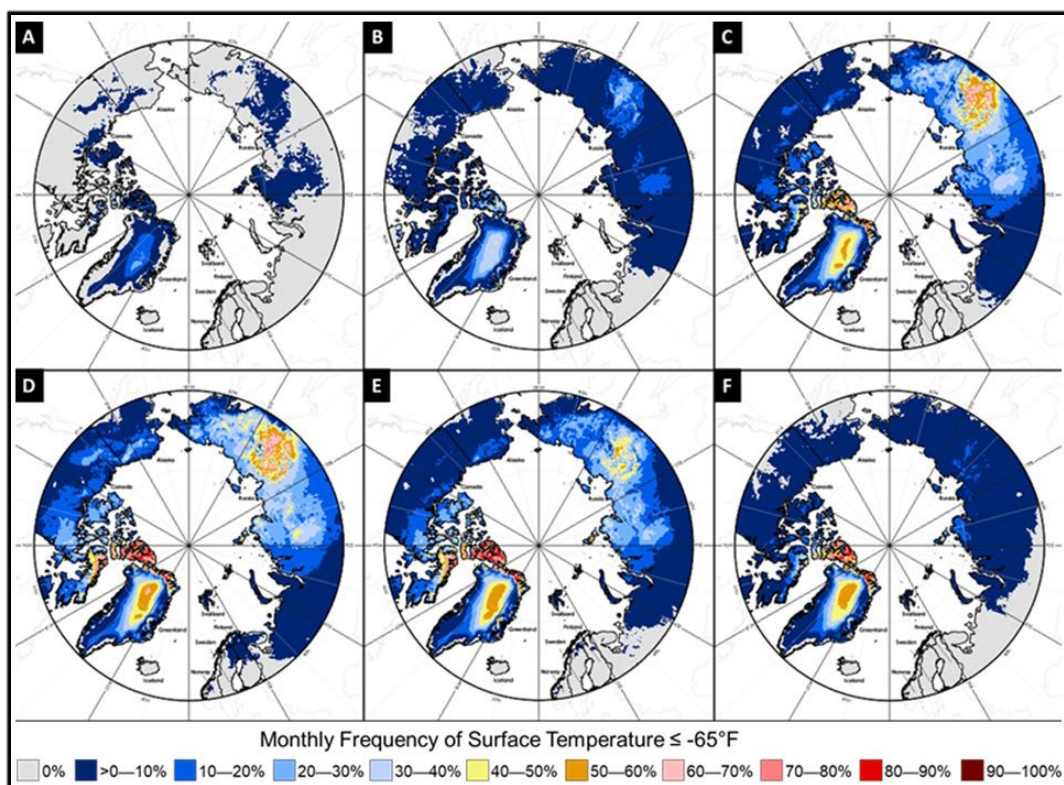


Table 32. Zonal statistics per location (within 60°N–90°N) for the lowest recorded surface temperature, defined as the six-month minimum (October–March), from hourly data from 1991 to 2020.

Location	Min (°F)	5th PCT (°F)	Mean (°F)	95th PCT (°F)	Max (°F)
Alaska	-76.7	-70.0	-56.4	-33.9	36.0
Russia	-99.9	-79.8	-66.0	-41.7	32.4
Finland	-56.0	-52.4	-37.7	-23.0	2.8
Sweden	-51.6	-48.3	-34.3	-19.4	-0.6
Norway	-53.5	-47.2	-27.5	11.2	37.9
Svalbard	-65.1	-60.1	-46.0	-24.5	0.4
Iceland	-38.3	-29.2	-15.7	9.6	33.1
Greenland	-84.6	-80.4	-64.0	-43.3	-9.8
Canada	-90.8	-78.8	-63.4	-49.2	-19.1

Figure 6. The monthly frequency of SKT between 60°N and 90°N latitude being recorded at or less than -40°F for October (A), November (B), December (C), January (D), February (E), and March (F), from hourly data between 1991 and 2020.



4.1.2 Two-meter Temperature (2T)

The average 2T across the entire 60°N to 90°N landmass between October and March was -5.4°F according to the 1991–2020 hourly measurements. The lowest temperatures were predominately in central Greenland (Figure 7). The average 2T in Scandinavia, western Russia, southern Canada, and southern Alaska did not get below -4°F , and nowhere averaged below -34°F when averaged for the portion of the country between 60°N to 90°N. The lowest average temperatures occurred in December and January (Figure 7).

The average of the lowest recorded 2T across the entire area of study between October and March for the period of record was -56.2°F , using hourly measurements from 1991 to 2020 (Table 33). Below -60°F 2T were recorded mostly in central Greenland and Russia, and below -40°F temperatures were recorded throughout the Arctic except for in most of Scandinavia (Figure 8; Table 34). The extreme temperatures were lowest in February and January (Figure 8).

Figure 7. Monthly average 2 m temperature (2T) between 60°N and 90°N for October (A), November (B), December (C), January (D), February (E), and March (F), from hourly data between 1991 and 2020.

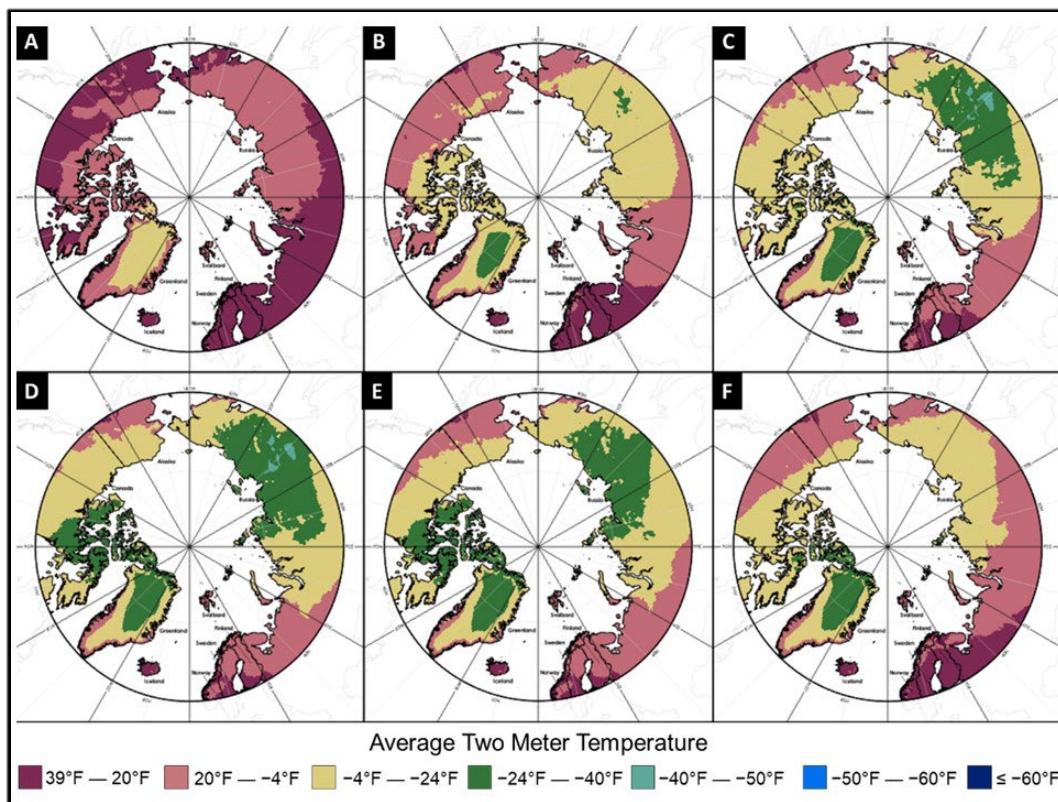


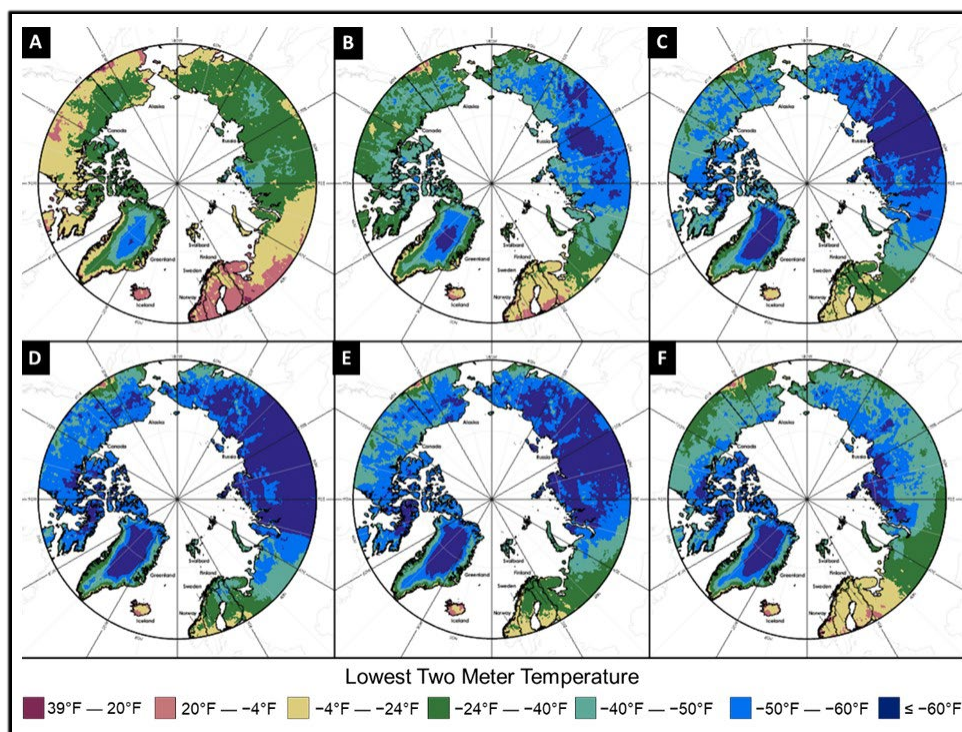
Table 33. Zonal statistics per location (within 60°N and 90°N) for the average 2T, according to hourly data from October–March 1991–2020.

Location	Min (°F)	5th PCT (°F)	Mean (°F)	95th PCT (°F)	Max (°F)
Alaska	-10.7	-7.0	4.9	19.9	39.3
Russia	-29.1	-19.7	-4.7	20.2	30.3
Finland	16.2	18.2	23.7	29.9	35.7
Sweden	15.3	17.2	23.5	30.9	34.7
Norway	16.0	18.0	25.1	35.0	41.4
Svalbard	2.3	3.2	8.9	17.1	29.0
Iceland	17.9	23.5	28.8	34.4	39.0
Greenland	-33.4	-31.9	-16.4	7.9	27.3
Canada	-28.5	-20.2	-8.5	6.5	16.2

Table 34. Zonal statistics per location for the lowest recorded 2T, using hourly data from October–March, 1991–2020.

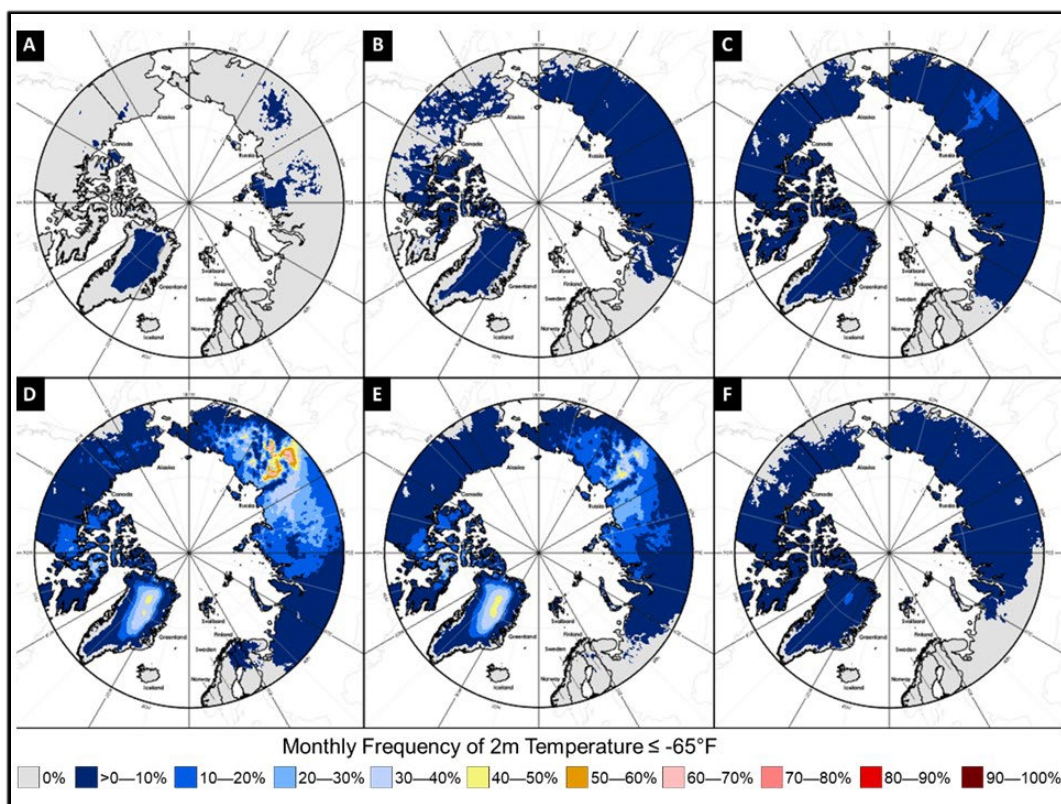
Location	Min (°F)	5th PCT (°F)	Mean (°F)	95th PCT (°F)	Max (°F)
Alaska	−74.0	−64.8	−50.7	−28.5	11.9
Russia	−84.6	−75.5	−60.7	−39.1	−4.2
Finland	−53.8	−49.8	−34.8	−20.9	−4.4
Sweden	−50.9	−44.5	−29.9	−16.1	0.8
Norway	−51.4	−43.8	−23.2	0.2	18.1
Svalbard	−51.6	−48.8	−38.9	−25.5	−9.7
Iceland	−21.9	−18.4	−6.9	6.4	13.2
Greenland	−82.4	−75.7	−56.2	−32.8	3.0
Canada	−78.5	−64.9	−55.2	−44.1	−25.8

Figure 8. Monthly lowest reanalysis hourly 2T between 60°N and 90°N latitude for October (A), November (B), December (C), January (D), February (E), and March (F), from hourly data between 1991 and 2020.



The average frequency of 2T being below −40°F between October through March was 5.02%, with the highest percentages occurring in central Greenland (32%) and Siberia (33%). The month of February had the highest frequencies of these extreme temperatures, reaching maximums of 41% in central Greenland. Figure 9 displays maps of 2T being below −40°F.

Figure 9. The monthly frequency of 2T between 60°N and 90°N latitude being recorded at or less than -40°F for October (A), November (B), December (C), January (D), February (E), and March (F), from hourly data between 1991 and 2020.



4.1.3 Soil Temperature, Level 1 (STL1)

The 0–7 cm average STL1 across the Arctic and Subarctic landmass averaged over October and March was 13.2°F according to the 1991–2020 hourly measurements. The lowest temperatures were predominately in central Greenland (Figure 10). The average winter STL1 in most of the Arctic did not get below -4°F except for Greenland, central Russia, and northern Canada (Table 35). However, the monthly average STL1 shows the coldest month is February (Figure 11). Table 36 shows minimum, maximum, mean, 5th and 95th percentile zonal statistics for STL1 per location.

Figure 10. Monthly average STL1 between 60°N and 90°N latitude for October (A), November (B), December (C), January (D), February (E), and March (F), from hourly data between 1991 and 2020.

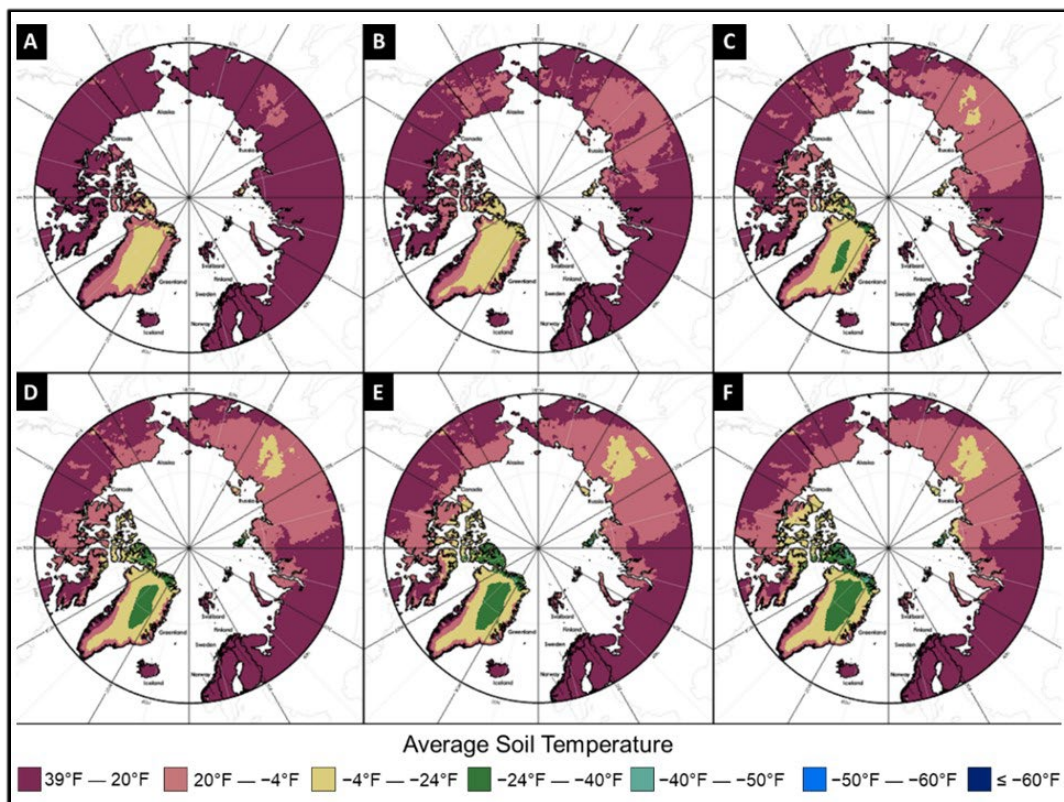


Table 35. Zonal statistics per location (within 60°N and 90°N) for the average STL1, using the hourly reanalysis data averaged over October to March, 1991–2020.

Location	Min (°F)	5th PCT (°F)	Mean (°F)	95th PCT (°F)	Max (°F)
Alaska	-5.3	13.7	22.2	30.1	44.4
Russia	-30.7	2.0	18.5	29.2	40.0
Finland	27.6	27.9	30.0	32.5	38.0
Sweden	27.8	28.7	30.3	32.5	36.5
Norway	27.1	29.8	32.7	39.3	45.4
Svalbard	13.6	17.8	23.9	29.5	33.3
Iceland	26.7	30.6	33.3	37.4	42.0
Greenland	-33.7	-25.4	-9.6	24.2	32.3
Canada	-30.6	-20.1	12.5	28.1	33.1

Figure 11. Monthly lowest hourly STL1 between 60°N and 90°N for October (A), November (B), December (C), January (D), February (E), and March (F), from hourly data between 1991 and 2020.

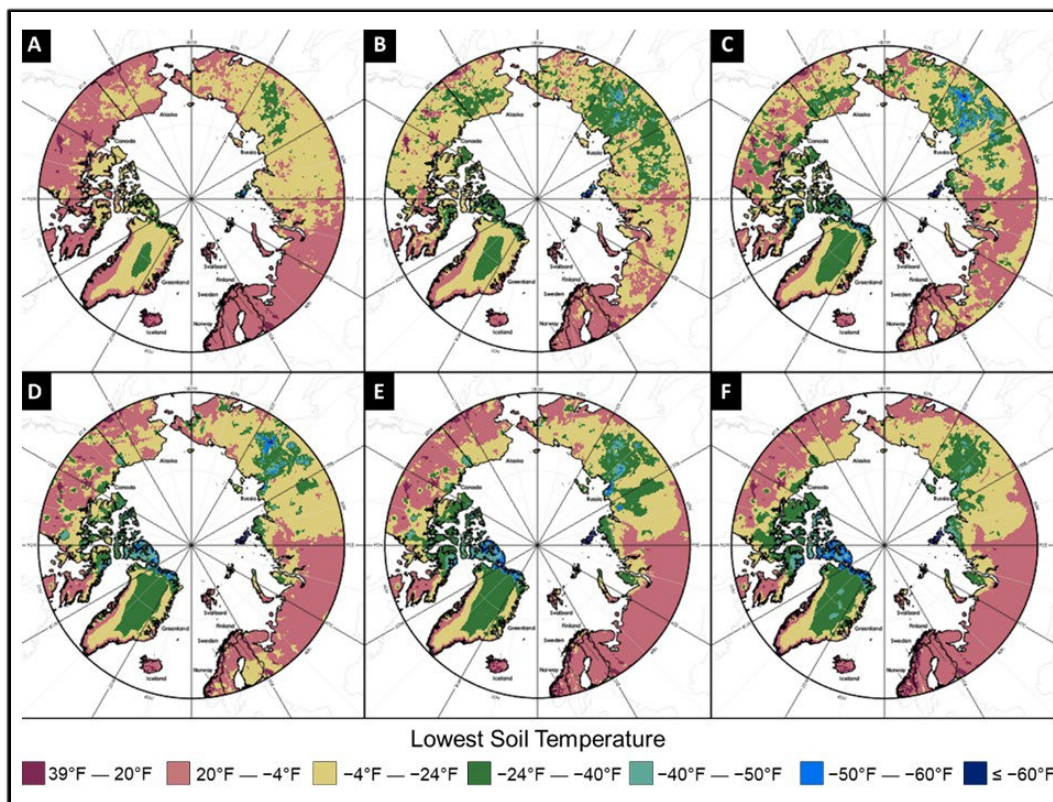


Table 36. Zonal statistics per location (within 60°N and 90°N) for the lowest recorded STL1, using hourly data from October–March, 1991–2020.

Location	Min (°F)	5th PCT (°F)	Mean (°F)	95th PCT (°F)	Max (°F)
Alaska	-49.9	-36.3	-18.7	4.0	38.3
Russia	-106.0	-45.3	-20.8	-0.1	33.0
Finland	-22.6	-19.9	-9.6	1.4	28.2
Sweden	-20.0	-12.2	-4.3	4.9	14.8
Norway	-21.1	-13.1	1.7	25.2	38.7
Svalbard	-29.5	-17.7	-4.4	5.6	13.7
Iceland	-2.2	2.9	13.0	26.5	36.4
Greenland	-69.7	-51.0	-28.4	-1.4	19.5
Canada	-69.5	-53.6	-27.0	-0.6	29.0

4.1.4 Wind Chill Temperature (WCT)

The average winter WCT across the Arctic and Subarctic landmass between October and March was -23.9°F using the 1991–2020 hourly measurements (Table 37). The lowest temperatures were predominately in central Greenland, northern Canada, and Russia, and January was the coldest month from these average temperatures (Figure 12). The average winter (October–March) WCT in Scandinavia, western Russia, southern Canada, and southern Alaska did not get below -4°F (Table 37).

Table 37. Zonal statistics per location (within 60°N and 90°N) for the average WCT, using hourly data averaged over October–March, 1991–2020.

Location	Min ($^{\circ}\text{F}$)	5th PCT ($^{\circ}\text{F}$)	Mean ($^{\circ}\text{F}$)	95th PCT ($^{\circ}\text{F}$)	Max ($^{\circ}\text{F}$)
Alaska	-32.5	-25.2	-8.8	9.4	37.1
Russia	-45.6	-38.0	-20.1	11.0	30.4
Finland	1.8	7.0	15.1	22.1	29.2
Sweden	2.0	5.3	14.3	23.5	27.9
Norway	0.9	4.3	14.1	28.2	38.6
Svalbard	-17.1	-14.7	-6.1	8.4	21.3
Iceland	-0.3	7.4	15.5	24.9	32.4
Greenland	-67.5	-65.9	-46.3	-12.3	19.5
Canada	-60.4	-50.2	-28.1	-4.0	6.2

The average lowest recorded WCT across the 60°N to 90°N landmass between October and March was -86.7°F using the 1991–2020 hourly derived data. Below -60°F surface temperatures were recorded across the Arctic, with Scandinavia being the highest region and Greenland being the lowest (Figure 13). The coldest month from these extreme values was February (Figure 12). Most of Greenland as well as northern Russia and Canada had experienced below -100°F WCTs, while most other parts of the Arctic had lows between -60°F and -100°F (Table 38).

The average frequency from number of hours per winter of wind chill temperatures being below -65°F was 7.66% (averaged for the full 60°N – 90°N land mass), with the highest frequency being in northeastern Greenland during the month of February (75%) (Figure 14).

Figure 12. Monthly average WCT between 60° N and 90° N latitude for October (A), November (B), December (C), January (D), February (E), and March (F) from hourly data between 1991 and 2020.

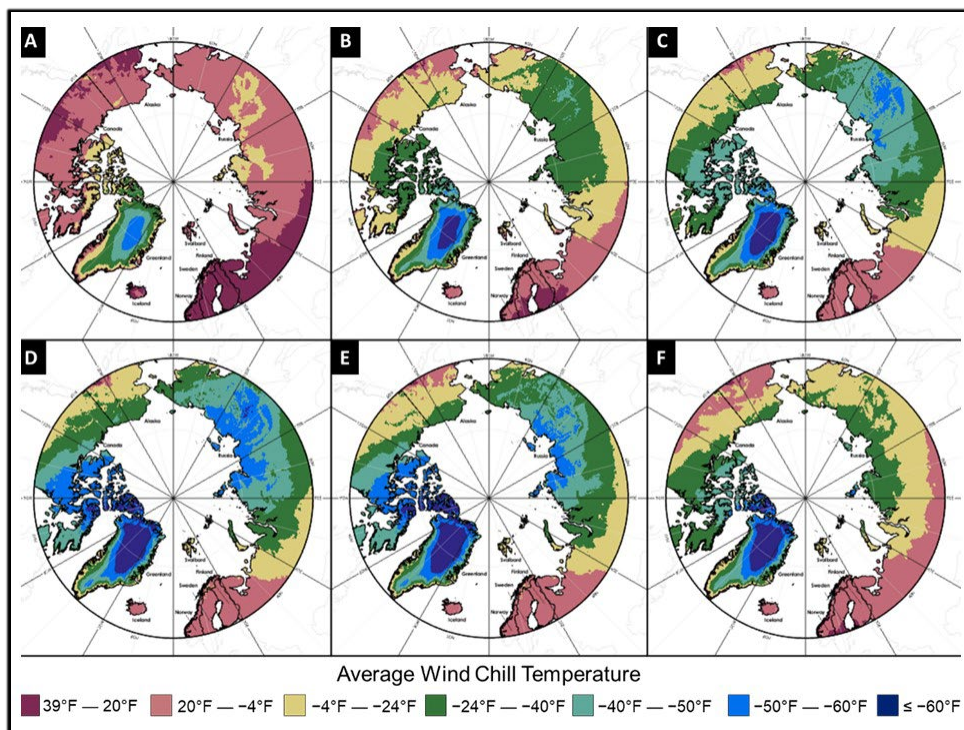


Figure 13. Monthly lowest WCT between 60° N and 90° N latitude for October (A), November (B), December (C), January (D), February (E), and March (F) from hourly data between 1991 and 2020.

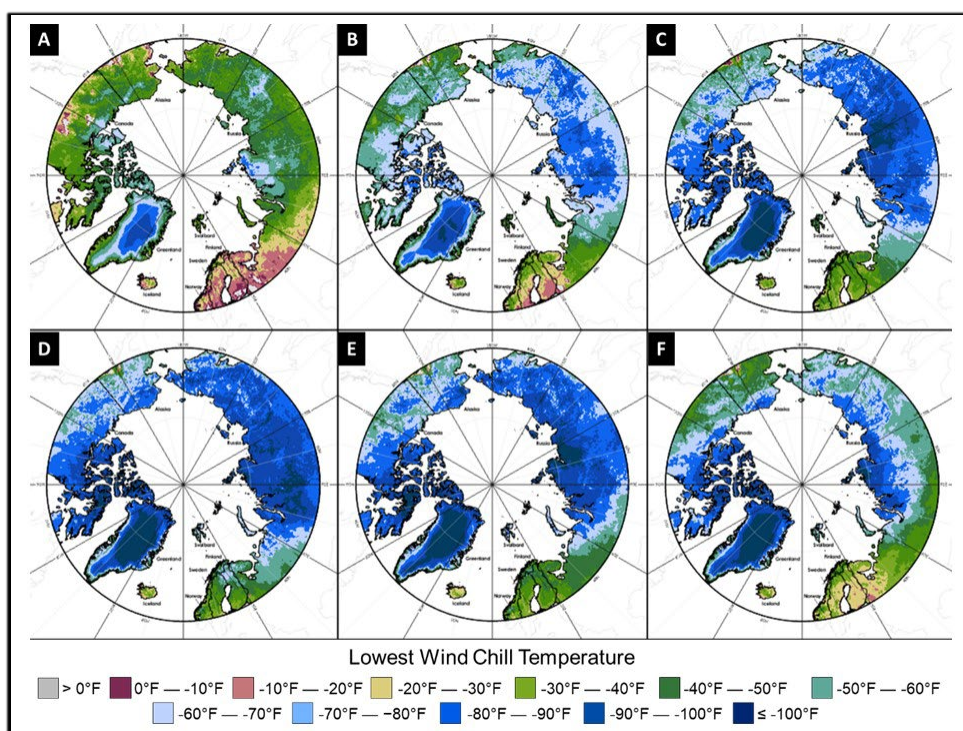
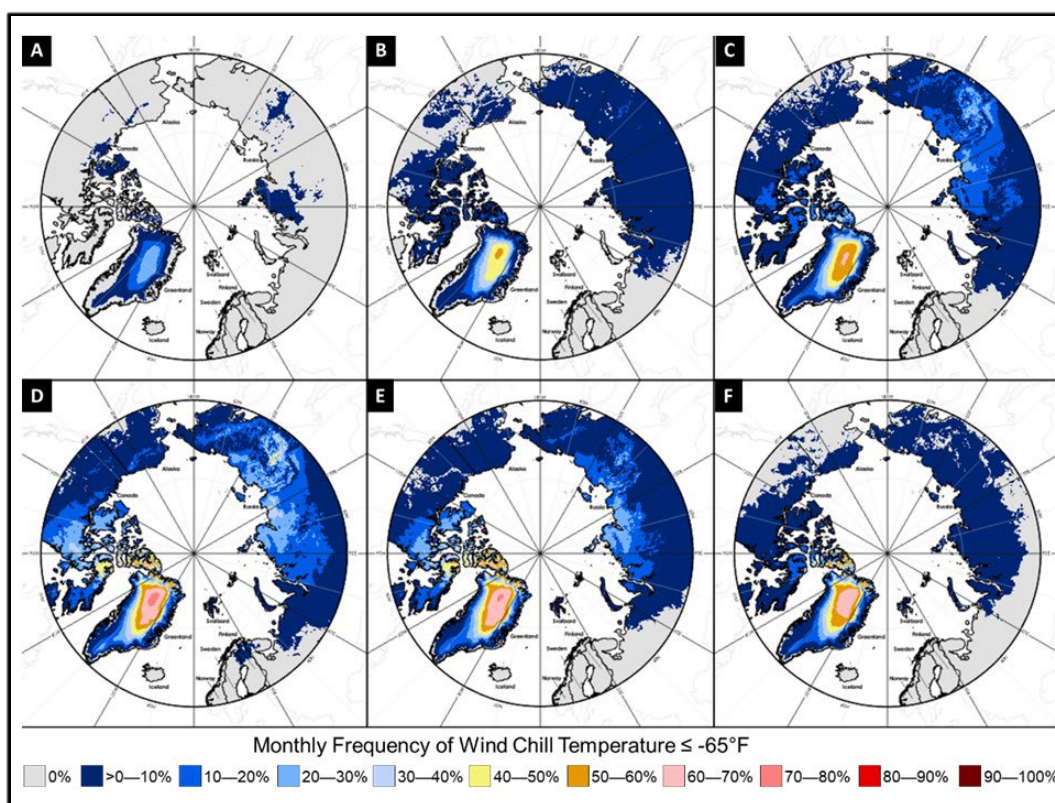


Table 38. Zonal statistics per location (within 60°N and 90°N) for the lowest recorded WCT, using hourly data from October–March, 1991–2020.

Location	Min (°F)	5th PCT (°F)	Mean (°F)	95th PCT (°F)	Max (°F)
Alaska	-103.7	-93.5	-76.1	-52.8	0.3
Russia	-129.9	-105.1	-88.5	-61.0	-4.3
Finland	-80.6	-73.8	-55.4	-38.0	-13.1
Sweden	-72.6	-67.3	-48.6	-34.5	-12.1
Norway	-75.3	-68.0	-44.2	-3.3	15.1
Svalbard	-84.8	-81.9	-67.8	-45.9	-21.4
Iceland	-50.2	-47.1	-30.9	-3.6	4.0
Greenland	-121.0	-116.7	-96.5	-66.9	-11.7
Canada	-117.5	-103.9	-87.6	-69.3	-51.3

Figure 14. The monthly frequency of WCT between 60°N and 90°N latitude being recorded at or less than -65°F for October (A), November (B), December (C), January (D), February (E), and March (F) from hourly data between 1991 and 2020.



4.2 Wind

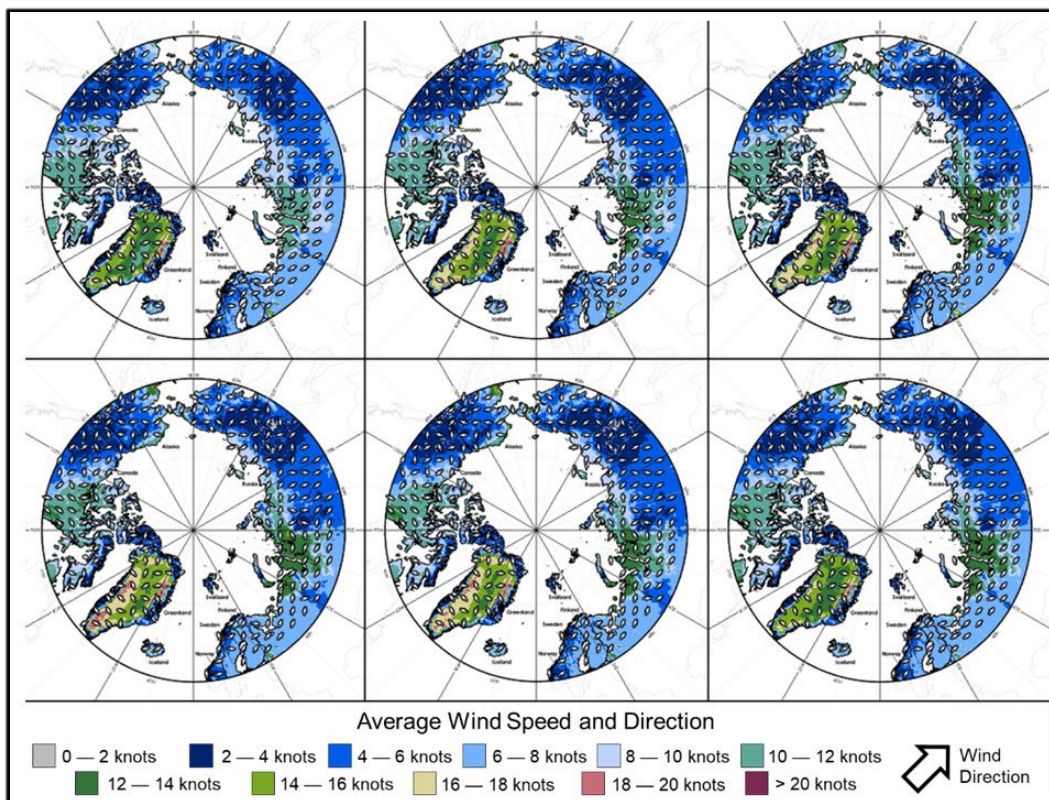
The average wind speed across the continental Arctic was 7.9 kn, with most of the higher wind speeds being in Greenland and some parts of

northern Canada and Russia (Table 39; Figure 15). The average wind speed peaked in January and was lowest in October (Figure 15).

Table 39. Zonal statistics per location (within 60°N and 90°N) for the average wind speed (knots), using hourly data from October–March, 1991–2020.

Location	Min (knots)	5th PCT (knots)	Mean (knots)	95th PCT (knots)	Max (knots)
Alaska	1.0	2.6	6.1	11.9	18.0
Russia	1.4	3.1	6.9	12.2	16.8
Finland	5.1	6.2	6.9	8.0	13.6
Sweden	3.1	4.4	6.1	7.1	11.2
Norway	2.1	3.3	6.0	10.9	17.1
Svalbard	2.9	3.7	8.3	13.4	18.6
Iceland	3.9	5.2	8.8	12.8	17.6
Greenland	0.4	3.0	12.1	17.3	21.8
Canada	0.7	2.7	7.4	11.5	18.2

Figure 15. Monthly average wind speed and direction between 60°N and 90°N latitude for October (*A*), November (*B*), December (*C*), January (*D*), February (*E*), and March (*F*), from hourly data between 1991 and 2020.

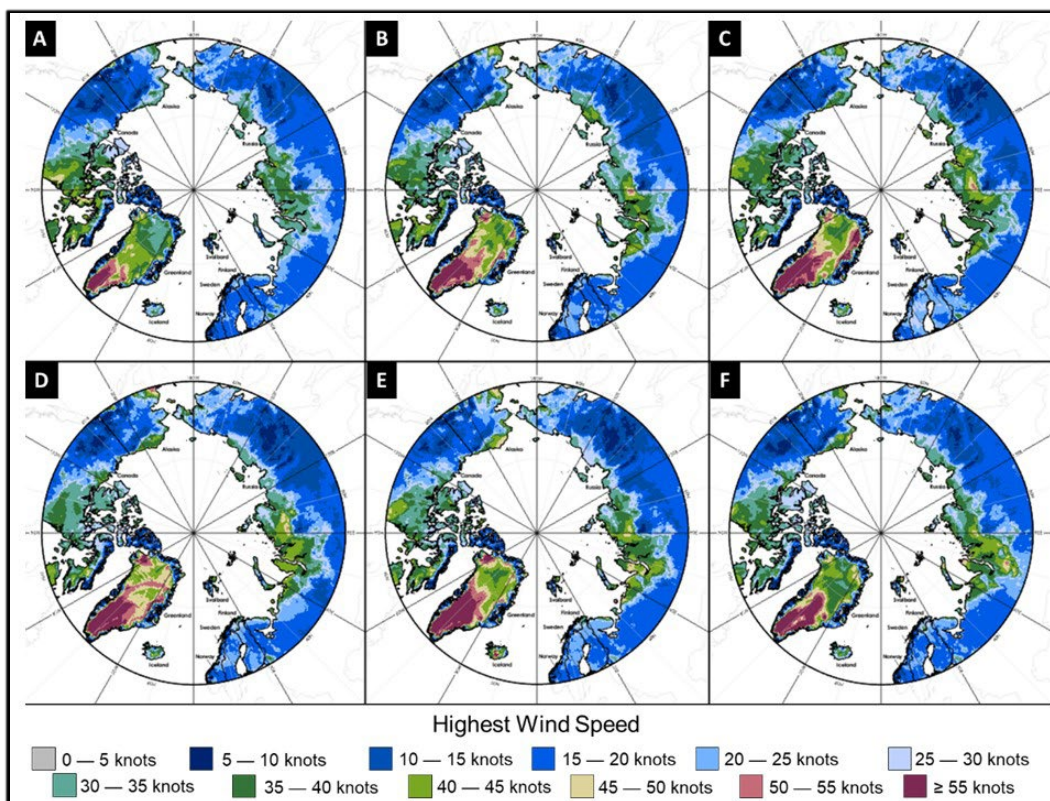


The highest full winter average wind speed averaged for all of the 60°N to 90°N landmass was 30.5 kn, with the highest wind speeds being in Greenland—especially southern Greenland and along the east coast—at 81 kn (Table 40; Figure 16).

Table 40. Zonal statistics per location (within 60°N and 90°N) for the highest recorded wind speed, using hourly data from October–March 1991–2020.

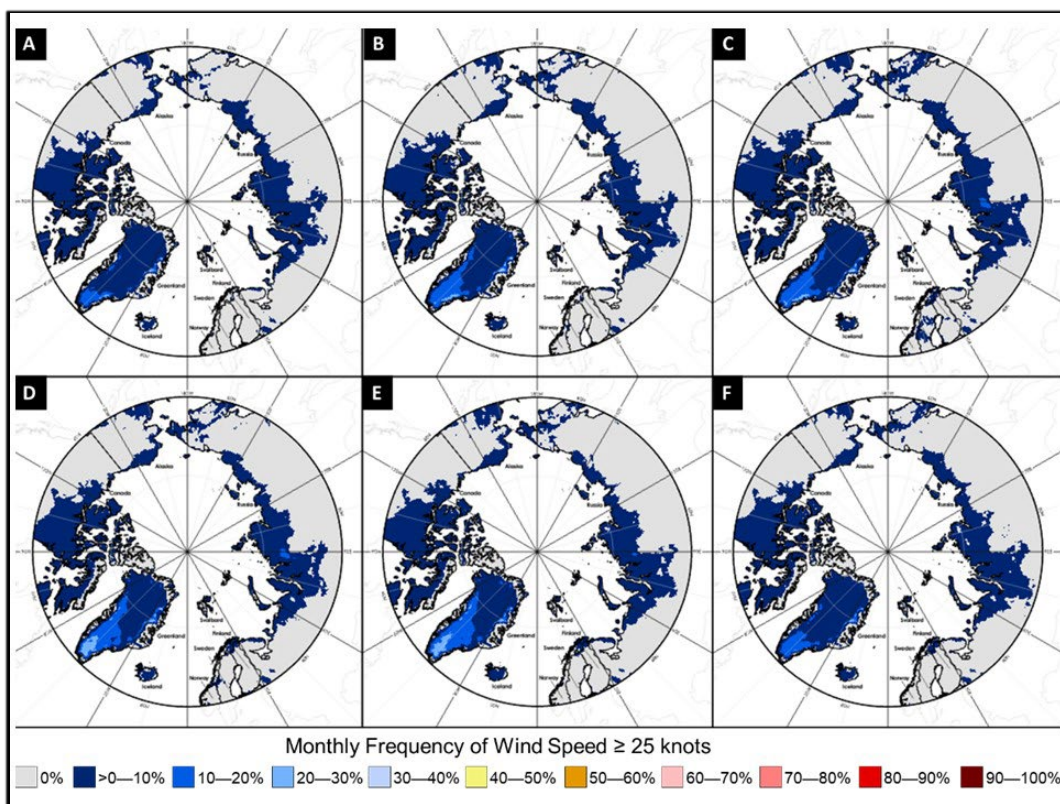
Location	Min (knots)	5th PCT (knots)	Max (knots)	95th PCT (knots)	Mean (knots)
Alaska	5.7	11.1	57.2	47.5	25.7
Russia	6.5	14.4	54.5	44.1	26.6
Finland	19.1	20.1	42.5	30.8	23.2
Sweden	16.1	20.1	38.1	28.7	24.1
Norway	10.1	15.2	57.6	42.9	25.6
Svalbard	14.4	18.5	57.9	52.0	34.0
Iceland	17.0	25.5	64.5	55.9	40.2
Greenland	2.8	15.7	81.0	67.0	45.5
Canada	4.8	11.4	55.2	44.1	29.5

Figure 16. Monthly highest wind speed between 60°N and 90°N latitude for October (A), November (B), December (C), January (D), February (E), and March (F), from hourly data between 1991 and 2020.



The average frequency of hourly wind speeds over the full winter being greater than 25 kn was 1.15%, with the highest frequency being in southern Greenland. The frequency of these extreme wind events peaked in February, with frequencies of around 22% in southern Greenland and 28% on the eastern coast of Greenland (Figure 17). The frequency of more extreme wind speed events, >80 kn, was near-zero across the area of study.

Figure 17. The monthly frequency of wind speed between 60°N and 90°N latitude being recorded at or higher than 25 °F for October (A), November (B), December (C), January (D), February (E), and March (F), from hourly data between 1991 and 2020.



4.3 Snow

4.3.1 Snow Depth (SD)

The winter average SD from October–March, 1991–2020 was 19.0 in. (Figure 18). The average highest SD across the Arctic was 44.4 in., with the highest SD being in the mountainous regions of Scandinavia, Russia, and Alaska (Figure 19). SD increased from October through March, as seen in Figure 19. Notably, the SD on ice sheets was not able to be calculated as the data was skewed by the ice and thus was masked out for Figures 18 and 19.

Figure 18. Monthly average SD between 60°N and 90°N latitude for October (A), November (B), December (C), January (D), February (E), and March (F), from hourly data between 1991 and 2020.

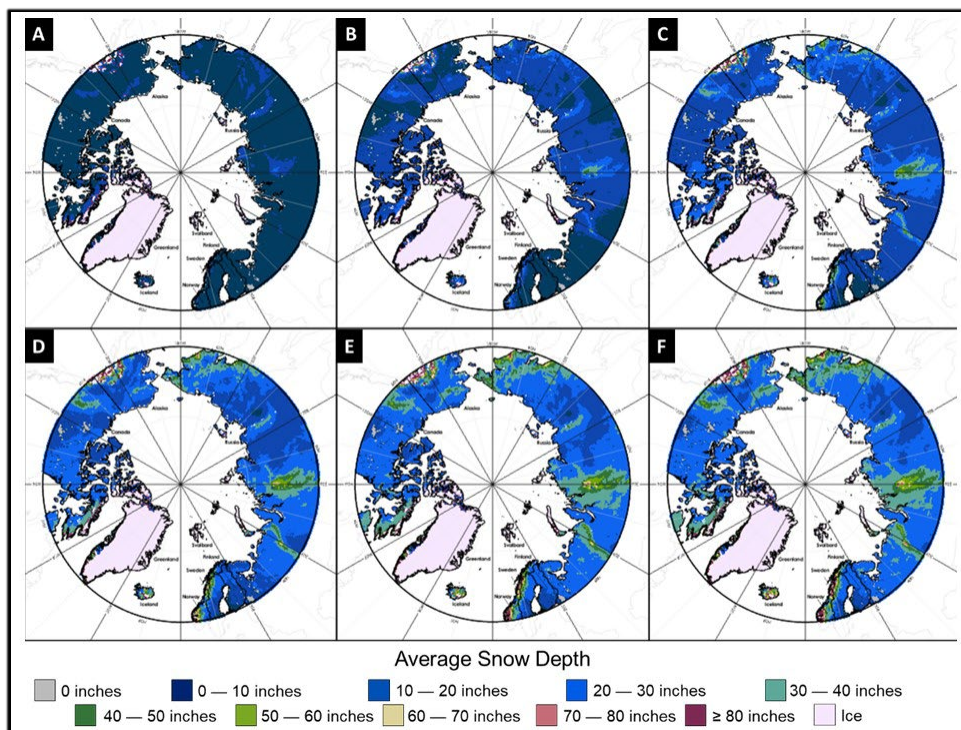
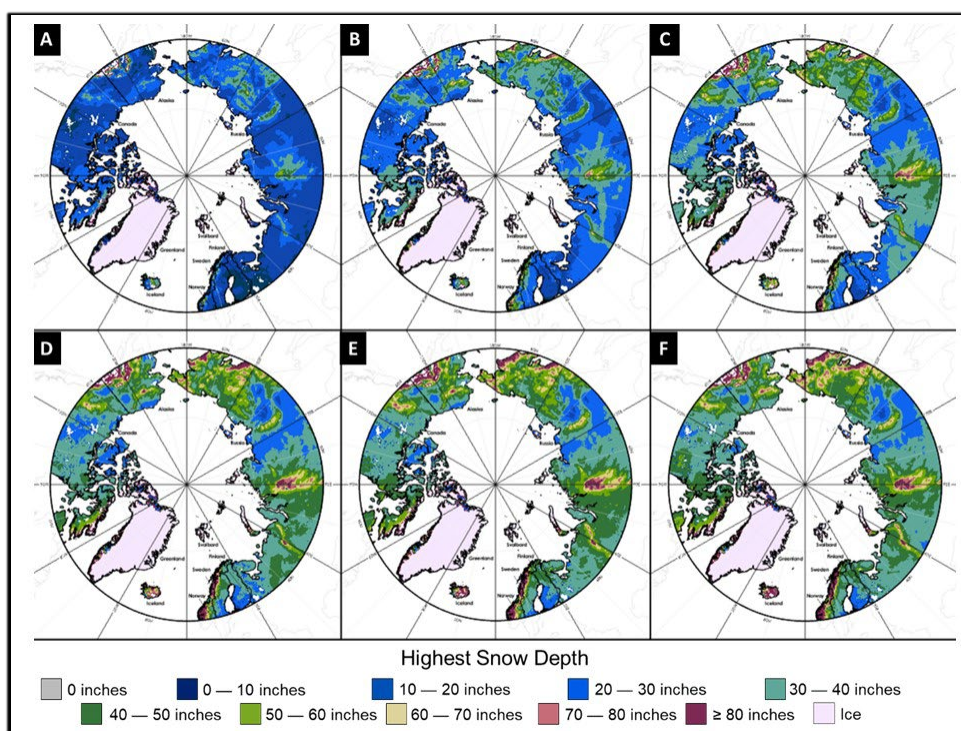


Figure 19. Monthly highest SD between 60°N and 90°N latitude for October (A), November (B), December (C), January (D), February (E), and March (F), from hourly data between 1991 and 2020.



4.3.2 Snow Load (SL)

The average SL calculated from October–March, 1991–2020 was 13.8 lb/ft², increasing from October through March (Figure 20). The average highest recorded SL across the Arctic was 42.9 lb/ft² with the highest snow load being in the mountainous regions of Scandinavia, Russia, and Alaska (Figure 21). Notably, the SL on ice sheets was not able to be calculated as the data was skewed by the ice and thus was masked out for Figures 20 and 21.

Figure 20. Monthly average SL between 60°N and 90°N latitude for October (A), November (B), December (C), January (D), February (E), and March (F), from hourly data between 1991 and 2020.

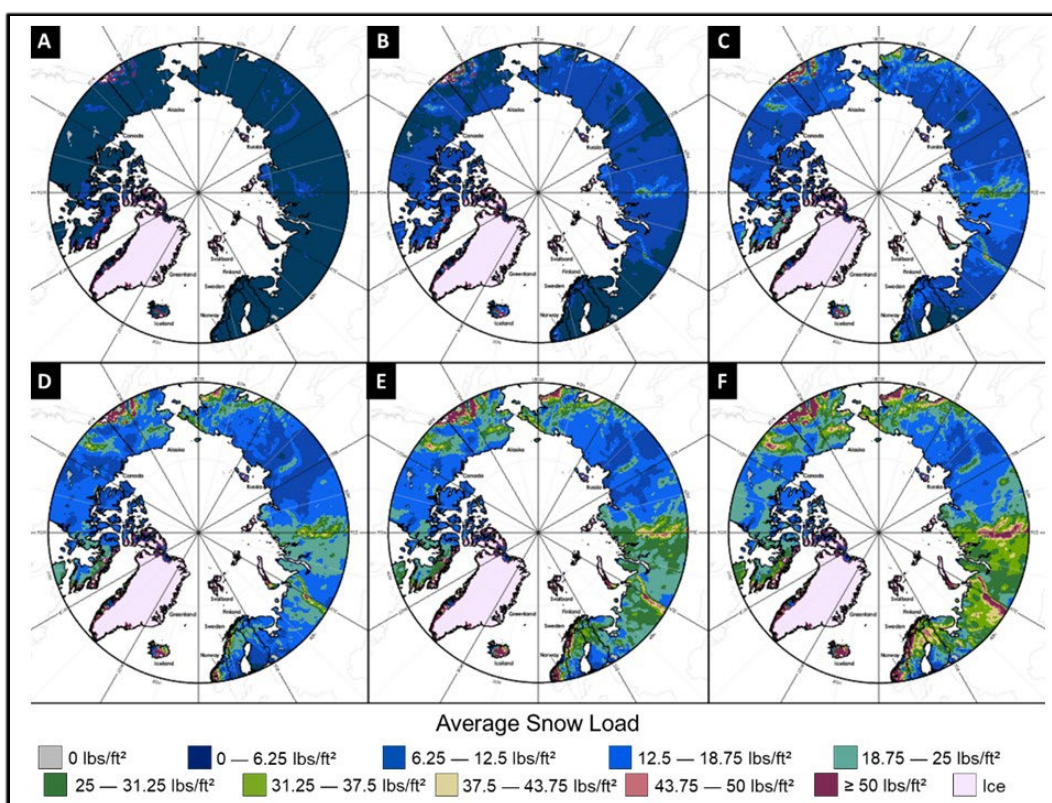
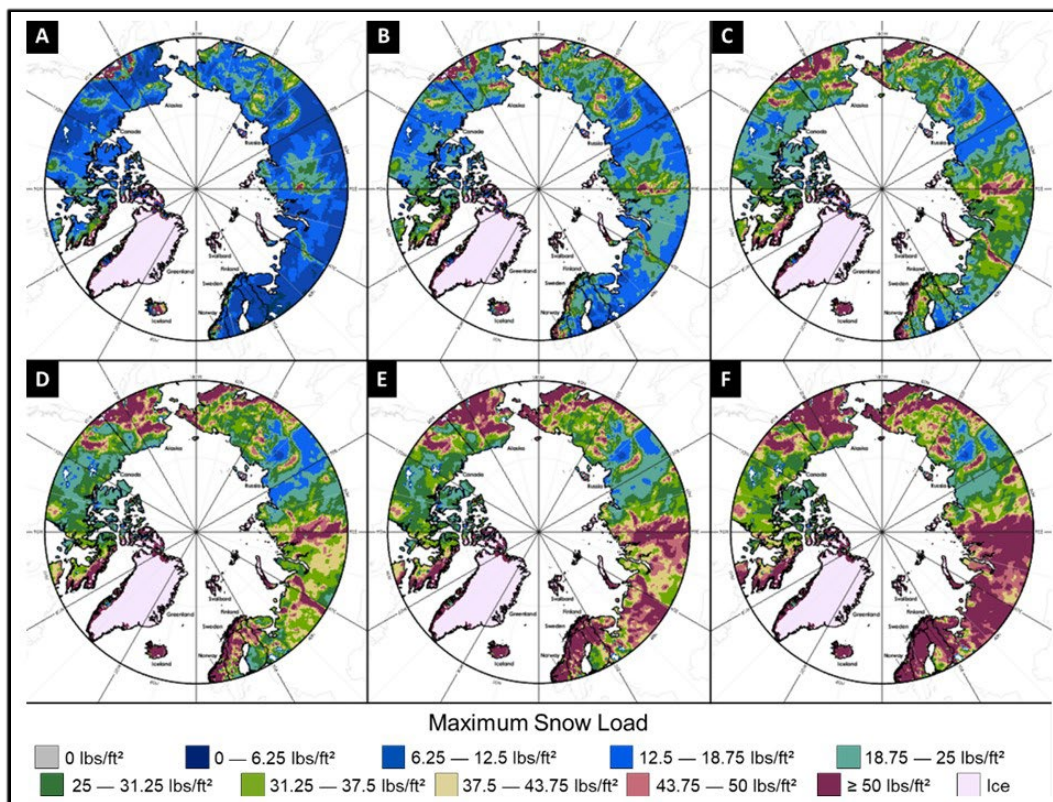


Figure 21. Monthly highest SL between 60°N and 90°N latitude for October (A), November (B), December (C), January (D), February (E), and March (F), from hourly data between 1991 and 2020.



5 Summary

The results of this project have produced a significant amount of information, data, and analyses in a relatively short time span. Over 40,000 maps of environmental Arctic and Subarctic conditions have been generated, validated, and archived for ready use (see Deliverables in Section 6.2). The compilation of almost 30 years of hourly reanalysis data for the circumpolar Arctic is archived and easily transferable to future users of these data. Additionally, these data are all parsed into key temporal intervals for operational use.

In reviewing the results of the generated maps for coincident extreme environmental parameters, we have concluded that while temperatures below -65°F or winds above 25 kn do occur, the frequency of such events is low ($<10\%$). In terms of operational planning purposes, specific materiel that is operable down to temperatures not quite as low as -65°F can still be deployed and operated in many locations. Average snow load rarely exceeds the 25 lb/ft² threshold, but can reach over 50 lb/ft² in many areas especially by February and March.

The comparisons between the ERA5 reanalysis data and the METAR observations show relatively good agreement and low standard deviation. There are a few anomalies where the ERA5 data and METAR air temperature observations deviate in a consistent matter, which is likely because of varying heights of weather instrumentation collecting observational data. The observed METAR wind speeds trend higher than the ERA5 wind speeds, which is likely because of the ability of the observational recorded data to capture extremes better than the gridded reanalysis dataset. In-depth analysis and discussion of these observations is detailed in the Special Report, *Development and Management of Arctic Zonal Characterization Products: Geospatial Database* (Barrett et al. 2025).

6 Discussion and Recommendations

6.1 Considerations for Soldiers

Cold environments are particularly difficult for soldiers because of the increased need for protective clothing and insulated shelters. Common atmospheric factors in cold environments include temperature, precipitation (snow and ice), fog, and wind. Low temperatures may also impair equipment operation, either temporarily or permanently, by changing the physical properties of materials. Hardening and brittleness of material, differential contraction of dissimilar materials, increased viscosity of lubricants, changes in electronic components, stiffening of shock mounts, and freezing of water are just a few of the potential problems. Ice accumulation on structures and equipment can cause damage and interfere with communication, visibility, and mobility. Aircraft such as helicopters and uncrewed aerial vehicles (UAVs) are especially vulnerable to icing, which can eliminate these key assets from operation. Accumulated snow also affects performance of electromagnetic sensors and additionally imposes a structural load on buildings, shelters, vehicles, and other equipment. There are human psychological and physiological challenges to operating in these extreme environments and the need for sustained training in these conditions cannot be overstated.

6.2 Deliverables and Future Needs

The deliverables of this project are ready for transition to a permanent location and use by the end-user community. To conclude this project in the 2025 fiscal year, the team at ERDC Cold Regions Research and Engineering Laboratory (CRREL) is working with the University of Alaska Fairbanks (UAF) Alaska Climate Research Center (ACRC) to transfer our compiled ERA5 dataset for hosting on their servers and long-term availability. UAF and CRREL are coproducing a landing page with description of the data on the ACRC's website (<https://akclimate.org/>) to allow for easy downloading access for end-users. Additionally, all generated maps will be transferred and accessible via a separate tab on the same website. ERDC-CRREL is developing a user interface (UI) to allow end-users to query their specified parameters and populate the corresponding map for download. For example, users will be able to select from the following options:

- *Region*: The landmass from 60°N to 90°N, which includes the entire Arctic landmass plus portions of Subarctic, 0°–90°W, 90°W–180°E, 180°E–90°E, and 90°E–0°.
- *Variable*: SKT, 2T, SD, WSPD, WCT, STL1, SL.
- *Metric*: average, extreme, frequency of extreme event (threshold specified), length of extreme event (threshold-specified).
- *Interval*: six hours, daily, weekly, biweekly, monthly, October–March Winter season.
- *Month*: October, November, December, January, February, and March.

Once this UI is available, end-users simply need to specify their desired parameters and a map will be generated and ready for download. This will allow users to download multiple maps and examine the potential for coincident extreme conditions.

In addition to this technology transfer effort, there are several options and potential pathways these datasets and maps can be integrated into end-user-defined tools. Although these efforts do not fall within the scope of the current project, they should be explored in future efforts. A useful effort could be to develop a standalone UI that generates maps on-demand as opposed to pulling from a static archive. This could additionally include the ability to overlap maps that narrow in on specific parameters that are of interest to view jointly. While the products developed for this project are specific to the Arctic region, the ERA5 dataset includes global coverage that could be applied to other regions of the world with regional observations, or globally. Lastly, the ability to zoom in to a resolution higher than 31 km on the maps is past the limitations of the datasets used in this study, but could be of interest in a future effort to allow for narrowing into smaller regions and viewing conditions at a higher fidelity than regionally.

Bibliography

- AMS (American Meteorological Society). 2024a. "Analysis." Glossary of Meteorology. <https://glossary.ametsoc.org/wiki/Analysis>.
- AMS (American Meteorological Society). 2024b. "Reanalysis." Glossary of Meteorology. <https://glossary.ametsoc.org/wiki/Reanalysis>.
- Barrett, E. P., B. Hoch, Z. J. Zody, M. L. Michaels, and R. T. Affleck. 2025. *Development and Management of Arctic Zonal Characterization Products: Geospatial Database*. ERDC/CRREL SR-25-2. US Army Cold Regions Research and Engineering Laboratory. <https://dx.doi.org/10.21079/11681/49991>.
- Curtis, J. n.d. *Snow Loads*. National Weather Service. <https://www.weather.gov/media/aik/articles/snowloads.pdf>.
- DAF (US Department of the Air Force). 2020. *Arctic Strategy: Ensuring a Stable Arctic Through Vigilance, Power Projection, Cooperation, and Preparation*. US Department of the Air Force. <https://www.af.mil/Portals/1/documents/2020SAF/July/ArcticStrategy.pdf>.
- DAF (US Department of the Air Force). 2022. *Flight Operations*. AFMAN 11-202, Volume 3. Headquarters, Department of the Air Force. https://static.e-publishing.af.mil/production/1/af_a3/publication/afman11-202v3/afman11-202v3.pdf.
- DAF (US Department of the Air Force). 2024. *C-17 Configuration and Mission Planning*. AFMAN 11-2C-17V3. Headquarters, Department of the Air Force. https://static.e-publishing.af.mil/production/1/af_a3/publication/afman11-2c-17v3add-a/afman11-2c-17v3add-a.pdf.
- DAF (US Department of the Air Force). n.d. "Country Climatologies." Air Force Combat Climatology Center. Accessed January 2025. <https://climate.af.mil/climatologies/country/>.
- DoD. 1997. *Department of Defense Handbook: Global Climatic Data for Developing Military Products*. MIL-HDBK-310. Department of Defense. <https://cvgstrategy.com/wp-content/uploads/2013/08/MIL-HDBK-310.pdf>.
- DoD. 2024. *2024 Arctic Strategy*. US Department of Defense. <https://media.defense.gov/2024/Jul/22/2003507411/-1/-1/0/DOD-ARCTIC-STRATEGY-2024.PDF>.
- DON (US Department of the Navy). 2019. *Strategic Outlook for the Arctic*. US Department of the Navy. https://www.c2f.usff.navy.mil/Portals/24/Documents/Navy_Strategic_Outlook_Arctic_Jan2019.pdf?ver=2019-09-26-105926-343.
- DOS (US Department of State). 2022. *National Strategy for the Arctic Region*. US Department of State. <https://bidenwhitehouse.archives.gov/wp-content/uploads/2022/10/National-Strategy-for-the-Arctic-Region.pdf>.

Earth Observing Laboratory. n.d. “Wind Direction Quick Reference.” National Center for Atmospheric Research. Accessed May 1, 2025. <https://www.eol.ucar.edu/content/wind-direction-quick-reference>.

ECMWF (European Center for Medium Range Weather Forecasting). 2018. “ERA5 Hourly Data on Single Levels from 1940 to Present.” Climate Data Store. <https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels?tab=overview>.

ECMWF (European Center for Medium Range Weather Forecasting). 2019. “ERA5-Land Monthly Averaged Data from 1950 to Present.” Climate Data Store. <https://cds.climate.copernicus.eu/datasets/reanalysis-era5-land-monthly-means?tab=overview>.

ECMWF (European Center for Medium Range Weather Forecasting). 2026. “ERA5: Data Documentation.” Documentation Centre. <https://confluence.ecmwf.int/display/CKB/ERA5%3A+data+documentation>.

ECMWF (European Center for Medium Range Weather Forecasting). n.d. “Climate Data Store.” Accessed 2025. <https://cds.climate.copernicus.eu/>.

Hersbach, H., B. Bell, P. Berrisford, S. Hirahara, A. Horányi, J. Muñoz-Sabater, J. Nicolas, et al. 2020. “The ERA5 Global Reanalysis.” *Quarterly Journal of the Royal Meteorological Society* 146 (730) 1999–2049. <https://doi.org/10.1002/qj.3803>.

HQDA (Headquarters, Department of the Army). 2020. *Research, Development, Test, and Evaluation of Materiel for Worldwide Use*. AR 70-38. Headquarters, Department of the Army. https://armypubs.army.mil/epubs/DR_pubs/DR_a/ARN30017-AR_70-38-000-WEB-1.pdf.

HQDA (Headquarters, Department of the Army). 2014. *Operator Manual for Extended Cold Weather Clothing System Generation III (ECWCS GENIII)*. TM 10-8415-236-10. Headquarters, Department of the Army. https://ciehub.info/ref/TM/10-8415-236-10_2014.pdf.

HQDA (Headquarters, Department of the Army). 2017. *Arctic and Cold Weather Maintenance Operations*. TM 4-33.31. Headquarters, Department of the Army.

HQDA (Headquarters, Department of the Army) and DAF (Department of the Air Force). 2021. *Weather Support for the US Army*. AR 115-10 / AFI 15-157. Headquarters, Department of the Army. https://armypubs.army.mil/epubs/DR_pubs/DR_a/ARN44730-AR_115-10-001-WEB-3.pdf.

HQDA (Headquarters, Department of the Army) and USMC. 2025. *Arctic and Extreme Cold Weather Operations*. ATP 3-90.96/MCTP 12-10E, Change 1. Headquarters, Department of the Army and Headquarters, United States Marine Corps. [https://www.marines.mil/Portals/1/Publications/MCTP%2012-10E%20\(SECURED\).pdf?ver=6UTIOU5xaA3ZkCUunrPJqQ%3D%3D](https://www.marines.mil/Portals/1/Publications/MCTP%2012-10E%20(SECURED).pdf?ver=6UTIOU5xaA3ZkCUunrPJqQ%3D%3D).

- ICAO (International Civil Aviation Organization). 2021. *Manual of Aeronautical Meteorological Practice, 13th Edition*. Doc 8896. International Civil Aviation Organization. <https://www2023.icao.int/airnavigation/METP/14th%20Meeting%20WAFS/METPWGMOG14%20IP09%20Att1-%20Doc.8896.update%20Amd%2081.pdf>.
- NCAR-NCL (National Center for Atmospheric Research–NCAR Command Language). 2019. “NCAR Command Language - Version 6.6.2.” https://www.ncl.ucar.edu/current_release.shtml.
- NCAR-NCL (National Center for Atmospheric Research–NCAR Command Language). n.d. “ncl_convert2nc.” NCL Tools. https://www.ncl.ucar.edu/Document/Tools/ncl_convert2nc.shtml.
- Nelson, C. A., M. Tew, G. Phetteplace, R. Schwerdt, A. Maarouf, R. Osczevski, M. Bluestein, et al. 2002. “Joint Development and Implementation by the United States and Canada of a New Wind Chill Temperature (WCT) Index.” The 15th Conference on Biometeorology and Aerobiology and the 16th International Congress of Biometeorology, Kansas City, MO, October 29. https://ams.confex.com/ams/15BioAero/techprogram/paper_48069.htm.
- NWS (National Weather Service). 2005. “WGRIB.” Climate Prediction Center. <https://www.cpc.ncep.noaa.gov/products/wesley/wgrib.html>.
- Pollard, Tom. 2023. “Python-metar.” Version 1.11.0. <https://github.com/python-metar/python-metar>.
- USAF (US Air Force). 2017. *14WS Support Analysis Request (SAR)*. 14 WS FORM 1. 14th Weather Service. https://climate.af.mil/static/docs/14ws_form_1_sar.pdf.
- WMO (World Meteorological Organization). 2017. *WMO Guidelines on the Calculation of Climate Normals*. WMO-No. 1203. World Meteorological Organization. <https://library.wmo.int/idurl/4/55797>.

Abbreviations

ABN DIV	Airborne Division
ACRC	Alaska Climate Research Center
AFMAN	Air Force Manual
AGM	Absorbed Glass Mat
AMMPS	Advanced Medium Mobile Power Sources
AMS	American Meteorological Society
AN-8	Antarctic Jet Fuel
AR	Army Regulation
ATP	Army Techniques Publications
CFS	Climate Forecast System
CIF	Central Issue Facility
CIG	Ceiling
CLP	Cleaner Lubricant Preservative
CRREL	Cold Regions Research and Engineering Laboratory
CSV	Comma separated value
DAF	US Department of the Air Force
DON	US Department of the Navy
DOS	US Department of State
ECMWF	European Centre for Medium-Range Weather Forecasts
ECWC	Extended Cold Weather Clothing
ERA5	European Center for Medium-Range Weather Forecasts Reanalysis v5
ERDC	Engineer Research and Development Center
FARP	Forward Arming and Refueling Point

14WS	14th Weather Squadron
GLP	General Purpose Lubricant
GMD	Grease, molybdenum disulfide
GRIB	Gridded binary
HQDA	Headquarters, Department of the Army
ICAO	International Civil Aviation Organization
IP	Interservice Publication
JP	Jet Propulsion
LAW	Lubricant Arctic Weight
LO	Lubrication order
LSA	Lubricant, Small Arms
LSAT	Lubricant, Weapon Semifluid (High-Load-Carrying Capacity)
MEP	Mobile Electric power
METAR	Meteorological Aerodrome Report
MTG	Military Tactical Generator
NCAR	National Center for Atmospheric Research
NCL	National Center for Atmospheric Research Command Language
netCDF	Network Common Data Format
NWP	Numerical weather prediction
NWS	National Weather Service
PCT	Percentile
PL	Preservative Lubricant
POL	Petroleum, oils, and lubricants
RBC	Rifle Bore Cleaner

RH	Relative humidity
SAR	Support Analysis Request
SAW	Squat Automatic Weapon
SD	Snow depth
SD-2	Degreasing Solvent
SFP	Single Fuel Policy
SKT	Skin temperature
SL	Snow load
SP	Special Purpose
STL1	Soil temperature, Level 1
SWE	Snow water equivalent
TM	Technical Manual
TQG	Tactical Quiet Generator
TSTMS	Thunderstorms
Turb	Turbulence
2T	2 m temperature
U	Wind Velocity, U-Component
UAF	University of Alaska Fairbanks
UAV	Uncrewed Aerial Vehicle
UI	User interface
USAF	United States Air Force
V	Wind velocity, V-Component
VIS	Visibility
WCT	Wind chill temperature
WDIR	Wind direction

WMO	World Meteorological Organization
WSPD	Wind speed
Xwind	Cross Wind

REPORT DOCUMENTATION PAGE

1. REPORT DATE March 2026		2. REPORT TYPE Final Technical Report		3. DATES COVERED	
				START DATE FY23	END DATE FY26
4. TITLE AND SUBTITLE Arctic and Subarctic Zonal Characterization and Operational Thresholding (AZCOT)					
5a. CONTRACT NUMBER		5b. GRANT NUMBER		5c. PROGRAM ELEMENT	
5d. PROJECT NUMBER		5e. TASK NUMBER		5f. WORK UNIT NUMBER	
6. AUTHOR(S) Rosa T. Affleck, Michelle L. Michaels, Brendon Hoch, Evan P. Barrett, Jenna N. Williams, Zachary J. Zody, Paige Toebben, and Mathew E. Hefner					
7. PERFORMING ORGANIZATION NAME(S) AND ADDRESS(ES) See reverse.				8. PERFORMING ORGANIZATION REPORT NUMBER ERDC/CRREL TR-26-5	
9. SPONSORING/MONITORING AGENCY NAME(S) AND ADDRESS(ES) US Army Corps of Engineers—Headquarters Washington, DC 20314-1000				10. SPONSOR/MONITOR'S ACRONYM(S) USACE	
				11. SPONSOR/MONITOR'S REPORT NUMBER(S)	
12. DISTRIBUTION/AVAILABILITY STATEMENT Distribution Statement A. Approved for public release: distribution is unlimited.					
13. SUPPLEMENTARY NOTES Project NH23-8375, "Arctic Zonal Characterization and Operational Thresholding (AZCOT)"					
14. ABSTRACT The US military develops and updates environmental parameters specified for the sustainment of operations throughout the world. These requirements are generally based on environmental data providing a baseline of temperature, wind, and precipitation expectations for each location. Observational data for Arctic regions is limited because of the remote and sparsely occupied geographical conditions. To address the need for updating these requirements, a 30-year analysis of meteorological conditions was conducted using a European Centre for Medium-Range Weather Forecasts (ECMWF) global reanalysis dataset over the Arctic and Subarctic region, defined by latitude 60°–90° North for this project. Raw hourly datasets were acquired, and the minimum temperatures, maximum wind speeds, maximum snow depths, and averages were determined over the period 1991–2020 between the months of October and March for each parameter. These were then visualized with geospatial analysis, producing a variety of maps designed to assist with the classification of parameters in Arctic zones of operation across a range of temporal resolutions. Finally, a review of operational limits for military equipment was conducted to match northern zones of operation with suitable capabilities dependent on environmental conditions.					
15. SUBJECT TERMS Arctic regions; Cold regions; Environment; Geospatial; Geospatial data; Meteorology; Military planning; Operations; Reanalysis; Thresholds; Weather forecasting					
16. SECURITY CLASSIFICATION OF:				17. LIMITATION OF ABSTRACT	
a. REPORT Unclassified	b. ABSTRACT Unclassified	c. THIS PAGE Unclassified	SAR		18. NUMBER OF PAGES 70
19a. NAME OF RESPONSIBLE PERSON Brendon Hoch				19b. TELEPHONE NUMBER (include area code) (603) 646-4295	

STANDARD FORM 298 (REV. 5/2020)

Prescribed by ANSI Std. Z39.18

PREVIOUS EDITION IS OBSOLETE.

7. PERFORMING ORGANIZATION NAMES AND ADDRESSES (concluded)

US Army Engineer Research and Development Center (ERDC)

Cold Regions Research and Engineering Laboratory (CRREL)

72 Lyme Road

Hanover, NH 03755-1290

Oak Ridge Associated Universities (ORAU)

Oak Ridge Institute for Science and Education (ORISE)

2902 Newmark Drive

PO Box 9,005

Champaign, IL 61,826-9005